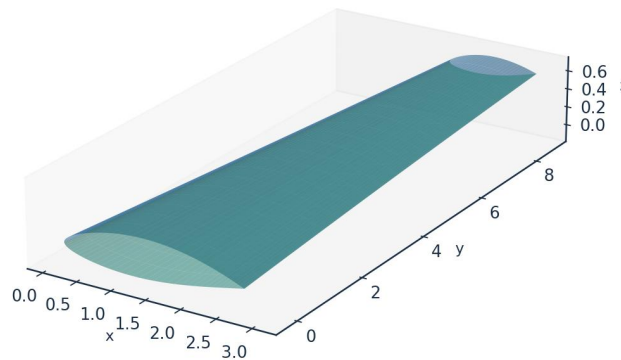


INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, $M=0.42$, $Re_{MAC} \approx 6.4 \times 10^6$):

Quantity	Predicted value
Upper-surface transition x_{tr}/c	0.554 (TS-natural)
Lower-surface transition x_{tr}/c	0.578 (TS-natural)
Mean laminar-flow extent	58.3 % of chord
Section lift coefficient C_l	0.517
Section profile drag C_d	43.8 counts
Viscous drag reduction vs fully-turbulent	50.4 %
Climb ($Tu=0.9$ %) transition x_{tr}/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number $Re_{\theta t}$ and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f , momentum thickness θ , shape factor H , intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e , with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at $M = 0.42$ against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below $M = 0.2$.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - $H^*(H)$, $Re_{\theta} C_f/2$ and $Re_{\theta} C_D$ - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns $H = 2.5914$ on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the $e^{\Lambda N}$ integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at $Re_{\theta} = 200$ against the accepted 200.5, a shape factor of 2.59129 and a wall shear parameter $f''(0) = 0.46960$. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 40 momentum thicknesses at 2.1 % free-stream turbulence to about 180 at 0.03 %.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) *Vortex-panel surface velocity (Kuethe & Chow)*

$$\frac{V_{t_i}}{U_{\infty}} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) *Kutta condition (sharp trailing edge)*

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) *Pressure coefficient (inviscid)*

$$C_p = 1 - \left(\frac{V}{U_{\infty}} \right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1+\beta}\right)\frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta \frac{dH^*}{dx} = 2C_D - H^* \frac{C_f}{2} - H^*(1-H) \frac{\theta}{U_e} \frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta \frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta \int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu ; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_{crit} ; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset uses the C1 criterion. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N , envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^x \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \geq N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U - c)(D^2 - \alpha^2) - U'']\hat{v} = \frac{1}{i\alpha Re_\theta}(D^2 - \alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma = -\alpha_i\theta = \frac{\omega_i}{c_g}, c_g = \frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin\Lambda \cos\Lambda \geq C_1$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf} = \int_{x_{c1}}^x \frac{\sigma(H_{rev})}{\theta} dx' \geq N_{crit}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx} = -(2 + H)\frac{\theta}{U_e}\frac{dU_e}{dx}, \theta\frac{dH^*}{dx} = 2C_D - H^*(1 - H)\frac{\theta}{U_e}\frac{dU_e}{dx}, C_f = 0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub} = \int_{x_s}^{x_r} \frac{\sigma(H_{rev}, Re_\theta)}{\theta} dx' = N_{crit}, \sigma(H_{rev}) \approx 0.0435$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^*$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left(\frac{U_{e,TE}}{U_\infty} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left(1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032 M_e^2 + 0.58 \left(\frac{T_w}{T_e} - 1 \right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e} \right)^{1+\omega}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e ⁿ N codes	RANS γ-Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e ⁿ N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mac_h	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.3)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

Wall y^+ target	~1 (reconstruction grid)
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Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	unity; validated on Schubauer-Skramstad and NLF(1)-0416 upper surface
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack $N = -8.43 - 2.4 \ln(Tu)$, clamped to $0.0008 \leq Tu \leq 0.0298$, anchored -0.50 (FITTED jointly on S&S and 86 aerofoil pts); 8.18 for $Tu < 0.08\%$
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_TS_max	0.1	upper Tu for the attached-flow TS branch	complement of the bypass gate
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical $Re_{\theta 2}$	Arnal et al.; facility-dependent: 150 fits Dagenhart (14.7%), 200 fits Boltz (18.4%)
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	θ_2/θ surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	0.5	offset on Mack's N_crit	unit conversion to the present amplification rates; set jointly on S&S and the 86 NLF(1)-0416 conditions
tu_hist	0.6	weight on the flow-history average of Tu, $Tu_{eff} = Tu_{avg}^w Tu_{local}^{(1-w)}$	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sep_floor	120.0	min separation-induced onset $Re_{\theta 2}$ (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
n_freq	40	frequencies carried by the e^N integration	discretisation parameter; onset moves <0.2% over 24-64
CF_N	0.0	separate amplification threshold for the crossflow branch (ablation path only)	0 = use the same N_crit as every other branch; exposed to test whether a roughness-seeded branch needs its own threshold, and it does not - see 06_validation/crossflow_criticals_summary.csv

Table 7. Calibration constants and their sources (input).

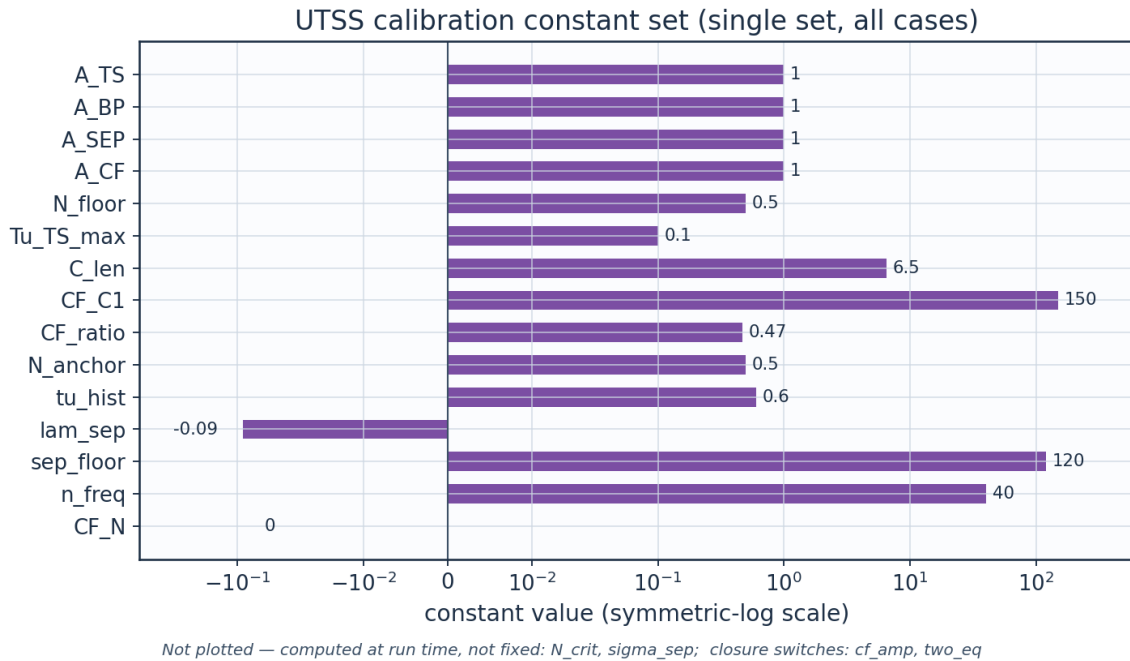


Fig. 8b. UTSS universal calibration constant set (*calibration_constants.csv*).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

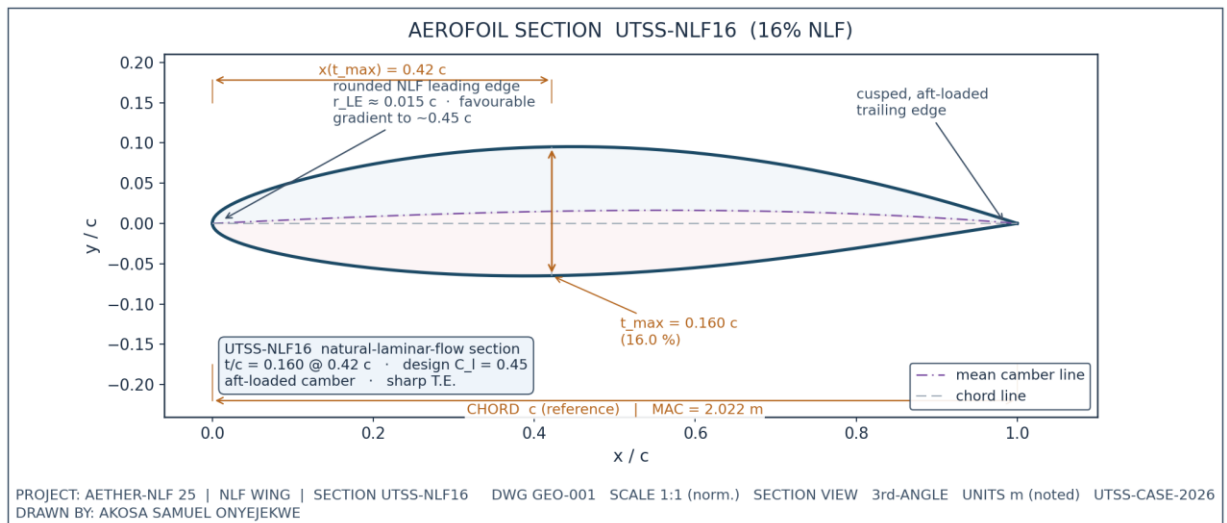


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

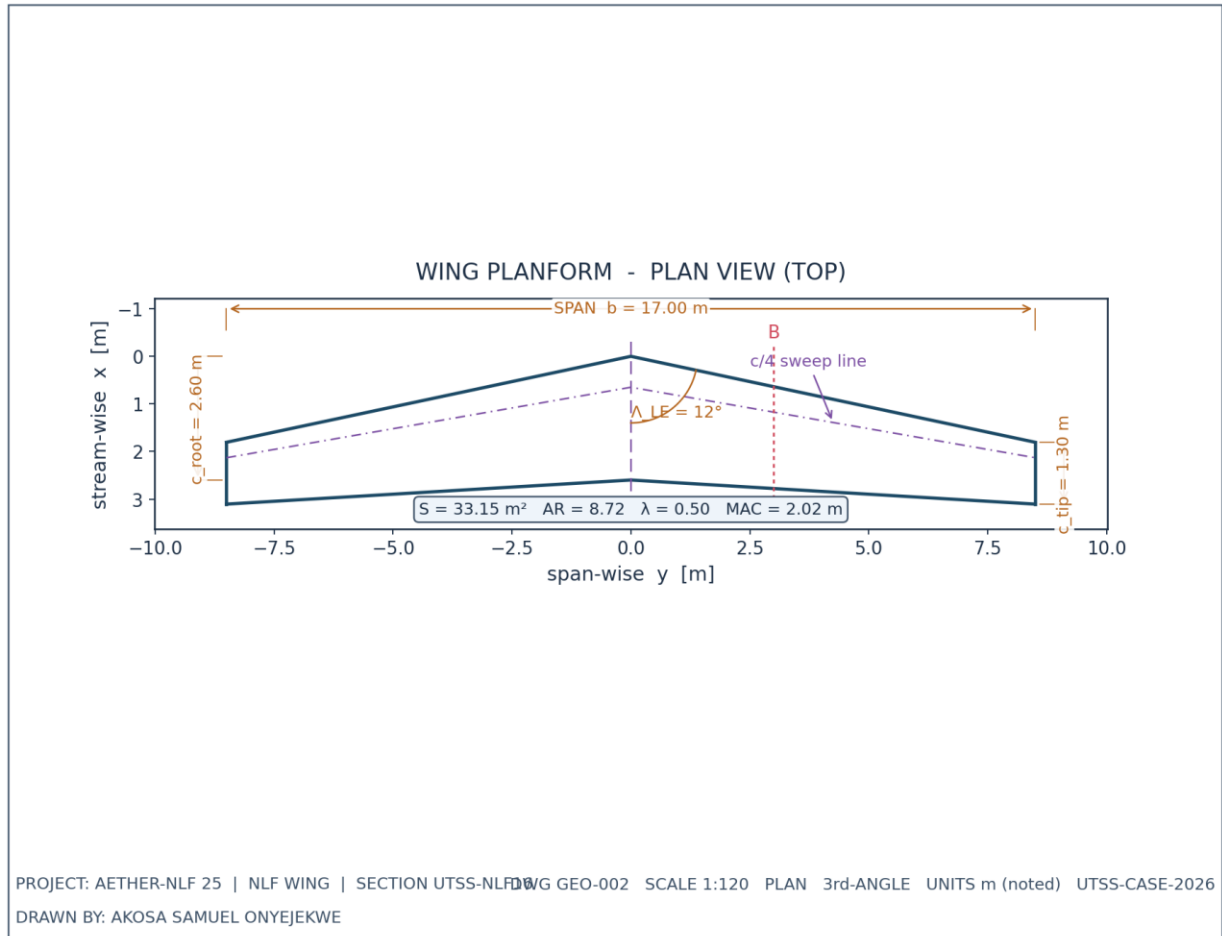


Fig. 2. Wing planform — plan (top) view, dimensioned.

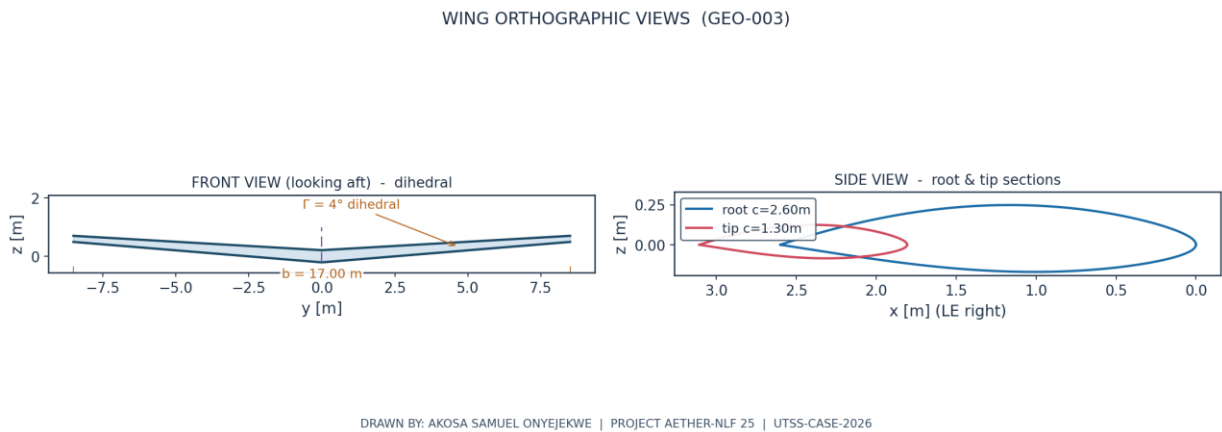
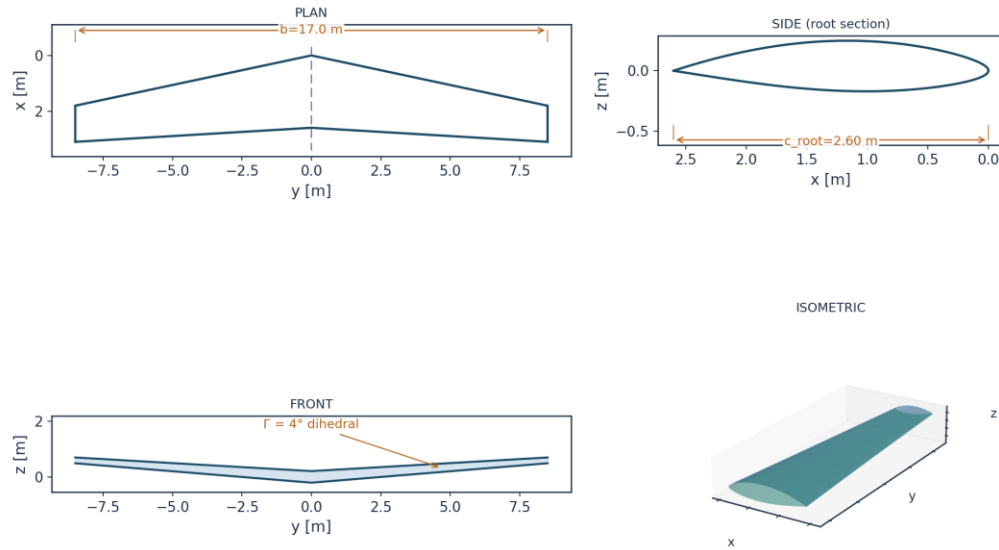


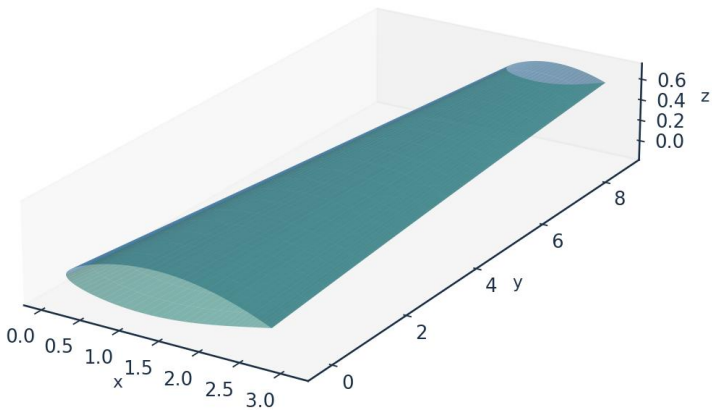
Fig. 3. Front and side orthographic views.



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 4. Full third-angle orthographic projection sheet.

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

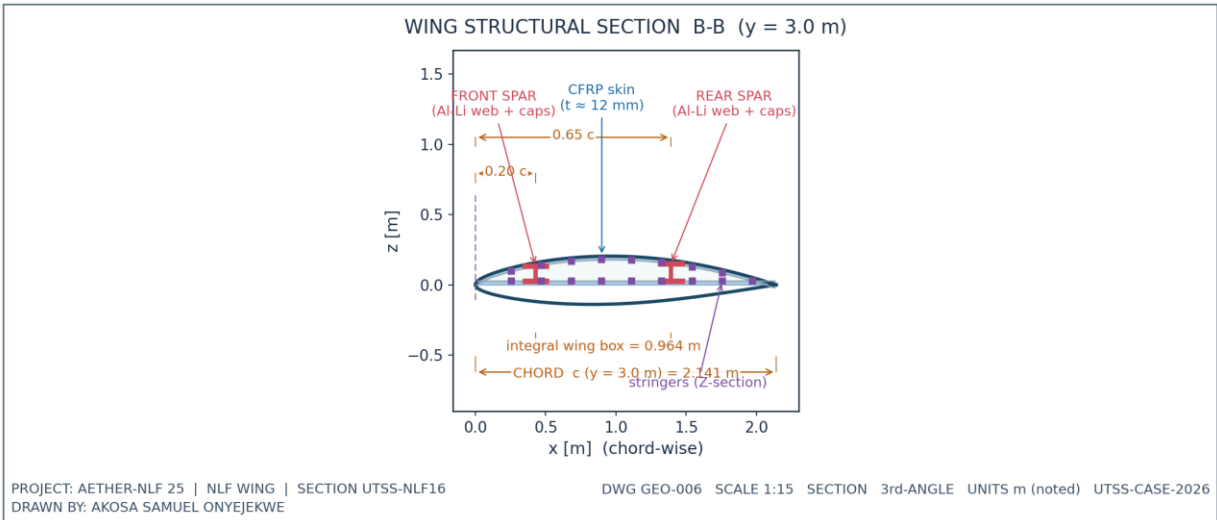


Fig. 6. Structural sectional view B-B.

7.1 Geometry data (from CSV)

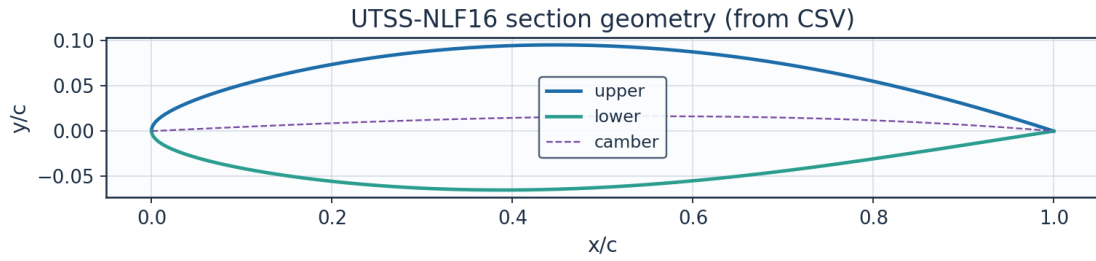


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

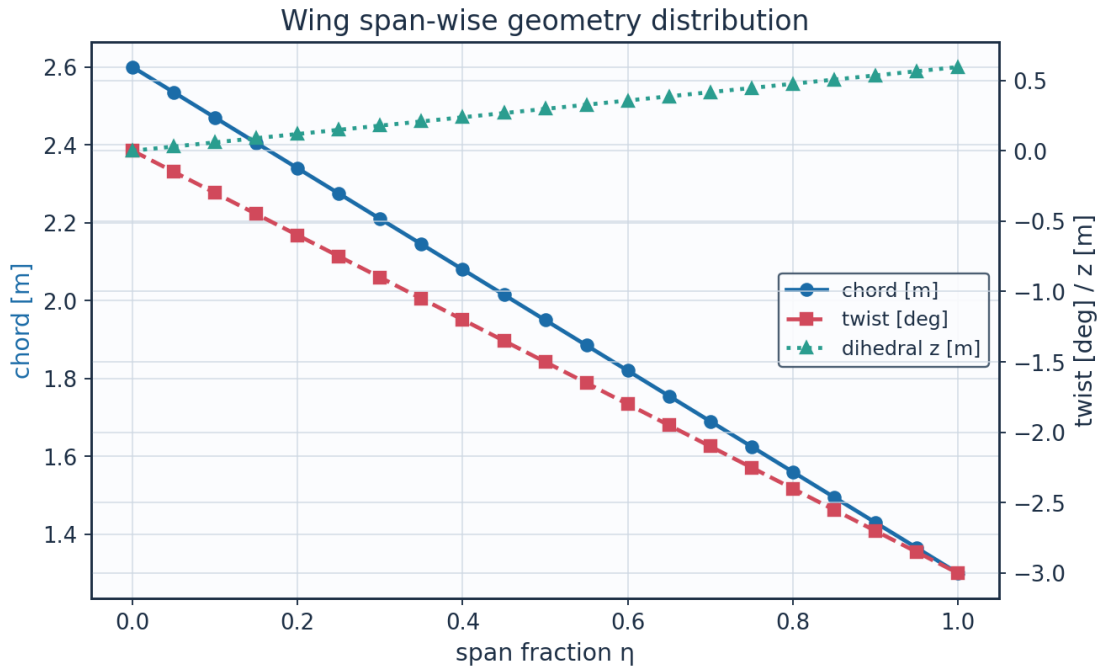


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012 995781
0.05	0.425	2.535	0.090336538 7097594	2.625336538 70976	-0.15	0.029718895 0759919	8040047.937 6708865
0.1	0.850000000 0000001	2.47	0.180673077 4195188	2.650673077 419519	-0.3	0.059437790 1519838	7833892.862 345993
0.15	1.275	2.405	0.271009616 1292782	2.676009616 1292783	-0.45	0.089156685 2279757	7627737.787 021098
0.2	1.700000000 0000002	2.34	0.361346154 8390376	2.701346154 8390376	- 0.600000000 0000001	0.118875580 3039677	7421582.711 696202
0.25	2.125	2.275	0.451682693 548797	2.726682693 548797	-0.75	0.148594475 3799596	7215427.636 371308
0.3	2.550000000 0000003	2.21	0.542019232 2585565	2.752019232 2585566	- 0.900000000 0000001	0.178313370 4559515	7009272.561 046414

0.35	2.975	2.145	0.632355770 9683159	2.777355770 968316	-1.05	0.208032265 5319434	6803117.485 721519
0.4	3.400000000 0000004	2.08	0.722692309 6780753	2.802692309 678076	- 1.200000000 0000002	0.237751160 6079354	6596962.410 396625
0.45	3.825	2.015	0.813028848 3878346	2.828028848 387835	-1.35	0.267470055 6839273	6390807.335 07173
0.5	4.25	1.95	0.903365387 097594	2.853365387 097594	-1.5	0.297188950 7599192	6184652.259 7468365
0.55	4.675000000 000001	1.885	0.993701925 8073536	2.878701925 807354	-1.65	0.326907845 8359112	5978497.184 421942
0.600000000 0000001	5.100000000 0000005	1.82	1.084038464 517113	2.904038464 5171127	- 1.800000000 0000005	0.356626740 9119031	5772342.109 097046
0.65	5.525	1.755	1.174375003 2268725	2.929375003 2268724	-1.95	0.386345635 987895	5566187.033 772152
0.700000000 0000001	5.95	1.69	1.264711541 9366315	2.954711541 936632	-2.1	0.416064531 0638869	5360031.958 447257
0.75	6.375	1.625	1.355048080 6463912	2.980048080 646391	-2.25	0.445783426 1398788	5153876.883 122363
0.8	6.800000000 000001	1.56	1.445384619 3561506	3.005384619 35615	- 2.400000000 0000004	0.475502321 2158709	4947721.807 797468
0.850000000 0000001	7.225	1.495	1.535721158 06591	3.030721158 06591	- 2.550000000 0000003	0.505221216 2918628	4741566.732 472573
0.9	7.65	1.43	1.626057696 7756692	3.056057696 775669	-2.7	0.534940111 3678547	4535411.657 14768
0.95	8.075000000 000001	1.365	1.716394235 485429	3.081394235 485429	-2.85	0.564659006 4438467	4329256.581 822785
1.0	8.5	1.3	1.806730774 195188	3.106730774 195188	-3.0	0.594377901 5198385	4123101.506 49789

Table 8. Wing planform stations (wing_planform.csv).

Lofted wing sections (from wing_sections_3d.csv)

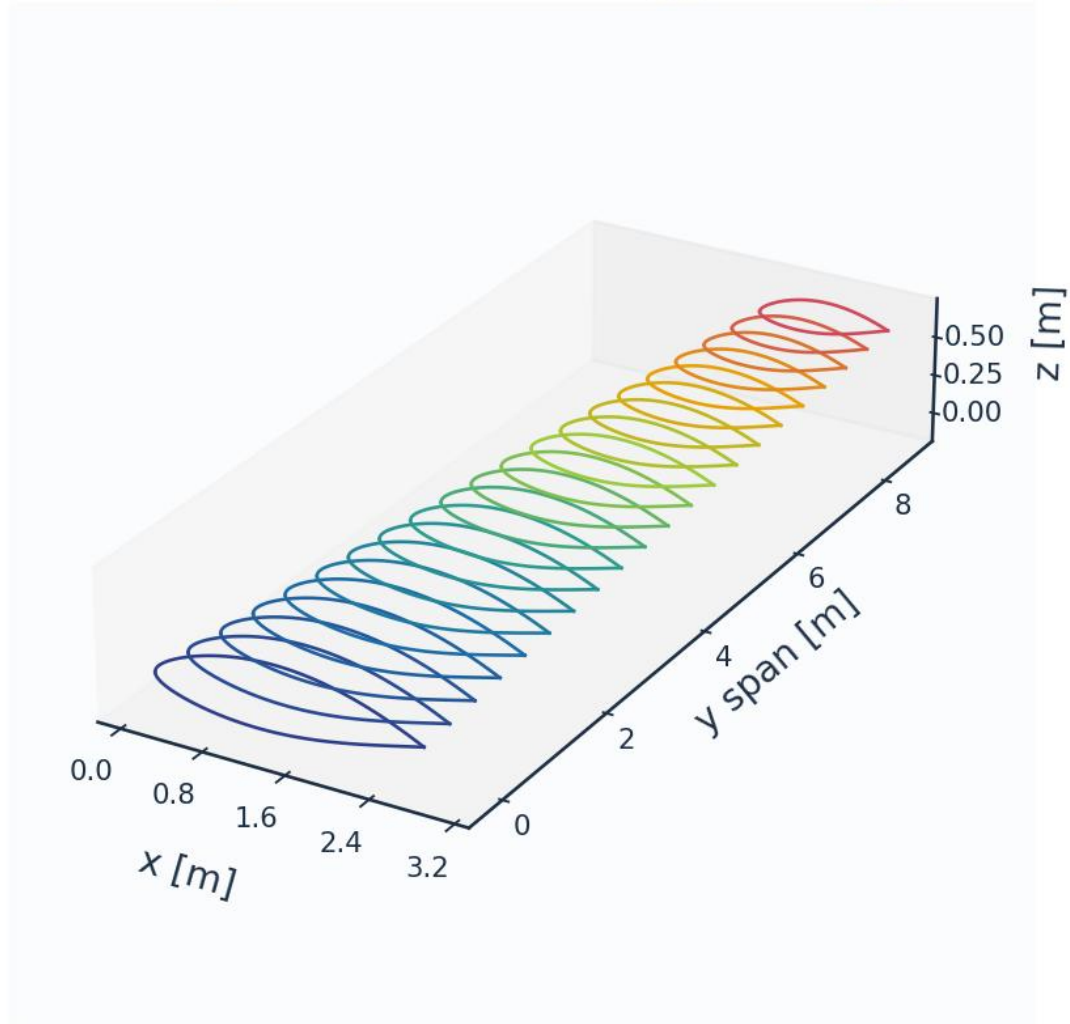


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A panel-count sweep bounds the discretisation sensitivity rather than demonstrating asymptotic convergence: from 120 to 480 surface panels the section drag spans 2.1 counts, 4.7 % of its mean, and above 180 panels it stays within about ± 0.5 count without tightening further. The residual wander is not a truncation error that refinement removes — it is set by which panel the transition point lands on, so it scales with the panel spacing at transition and is of the same order as the $\pm 0.025c$ bracket the aerofoil measurements themselves carry. The 260-panel grid is used throughout.

metric	value	unit
--------	-------	------

Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y_1	6.51	micron
Target wall y^+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_τ (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.5146538348994995	0.0042584577133717	0.538986194980472	
180	0.5161840825332155	0.0043773257315635	0.5433865690013859	2.791339639666557
260	0.5170864702757358	0.0043837763667638	0.5541908676652394	0.1473647518120291
360	0.5176435398599823	0.0044661036655082	0.5477820283105952	1.8779995113031636
480	0.5180064091537756	0.0044403358876658	0.5554577472375675	-
				0.5769632720663997

Table 10. Panel-count sensitivity study.

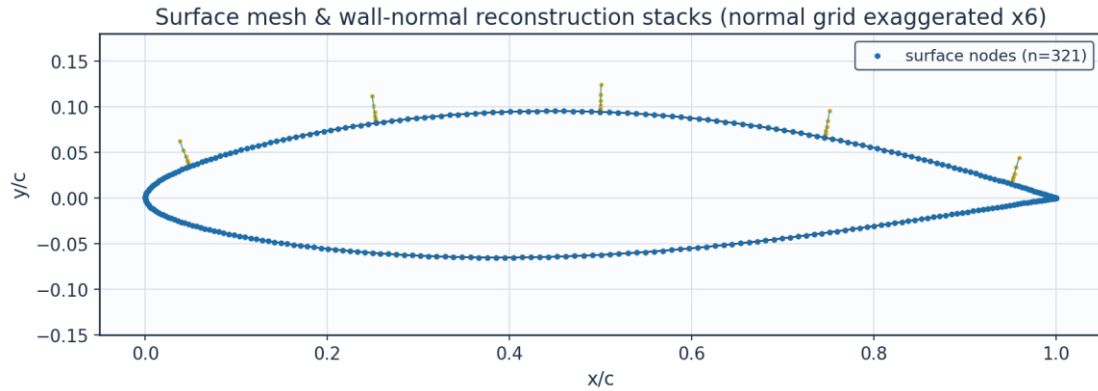


Fig. 9. Surface mesh and wall-normal stacks.

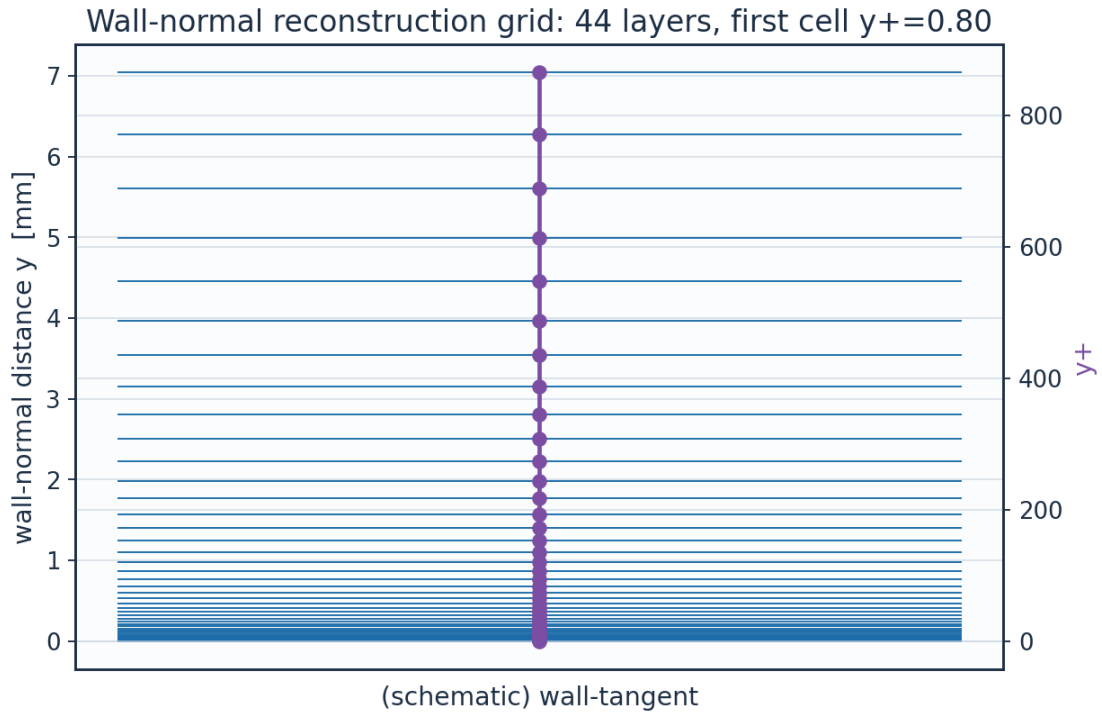


Fig. 10. Wall-normal reconstruction grid / y^+ .

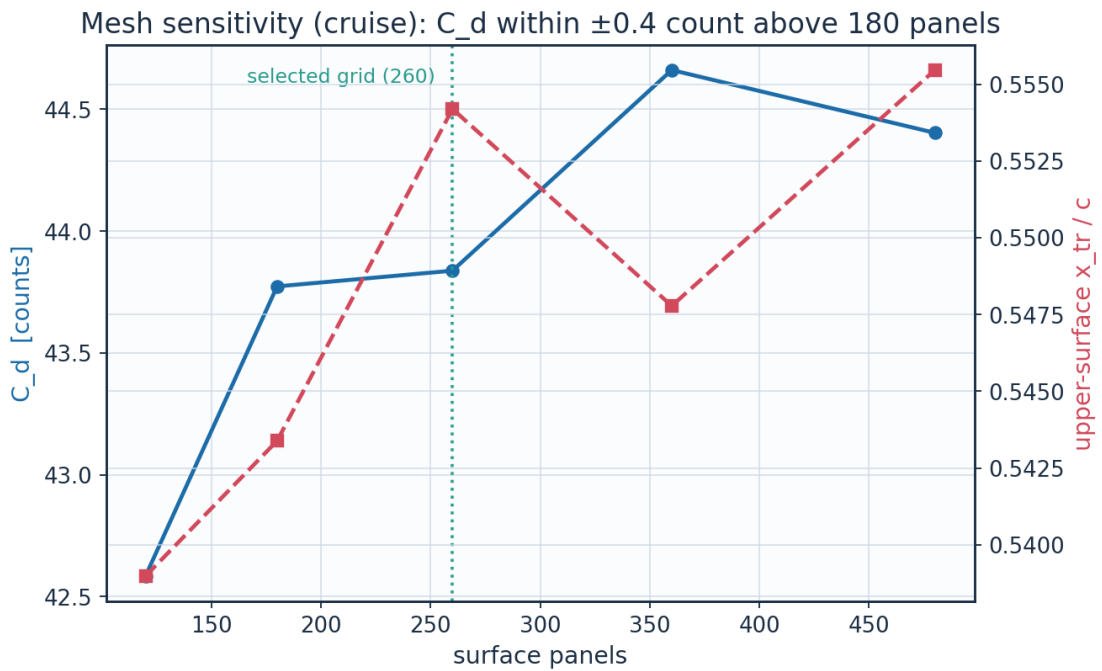


Fig. 11. Panel-count sensitivity of C_d and transition location.

layer	y_m	y_{plus}	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.1200000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.1199999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.579	0.554	3712000.0	1410.4	TS-natural	55.4	0.00021
CRUISE	lower	0.587	0.578	3766000.0	1270.7	TS-natural	57.8	0.00025
CLIMB	upper	0.047	0.025	506900.0	483.8	bypass	2.5	0.00014
CLIMB	lower	0.159	0.157	1706000.0	610.7	bypass	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv).

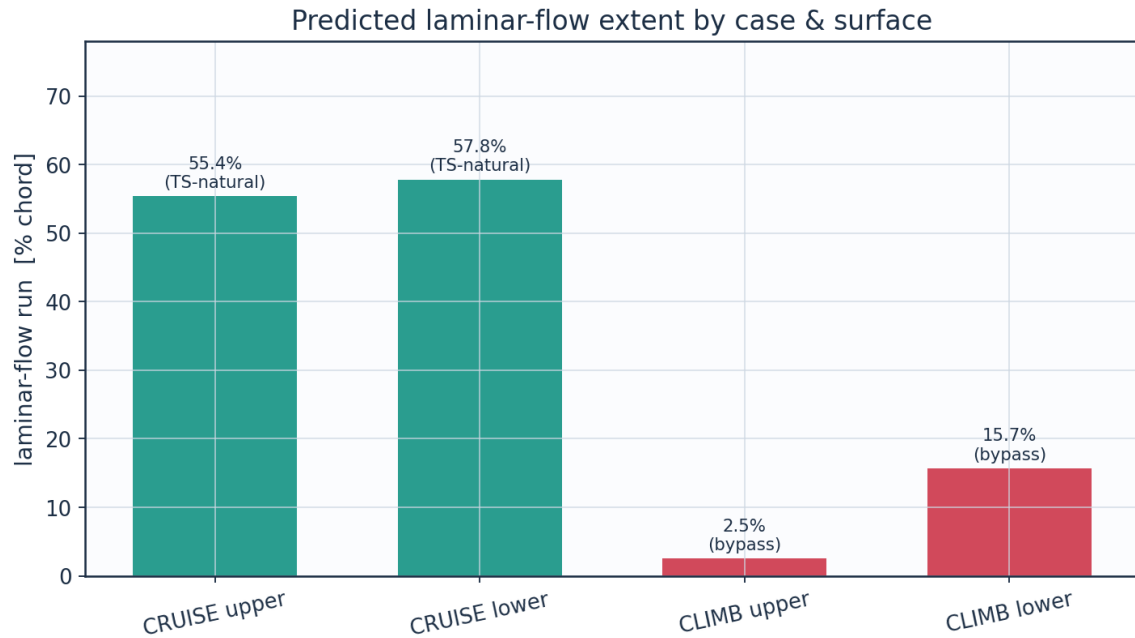


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00438	-
Section Cd (counts)	43.8	counts
Section L/D	118.0	-
Wing lift (3-D estimate, C _L = 0.90 C _l)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re _{MAC}	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	43.8	58.3
Fully turbulent (LE trip)	88.4	0.0
Viscous drag reduction	50.4	

Table 14. NLF vs fully-turbulent drag.

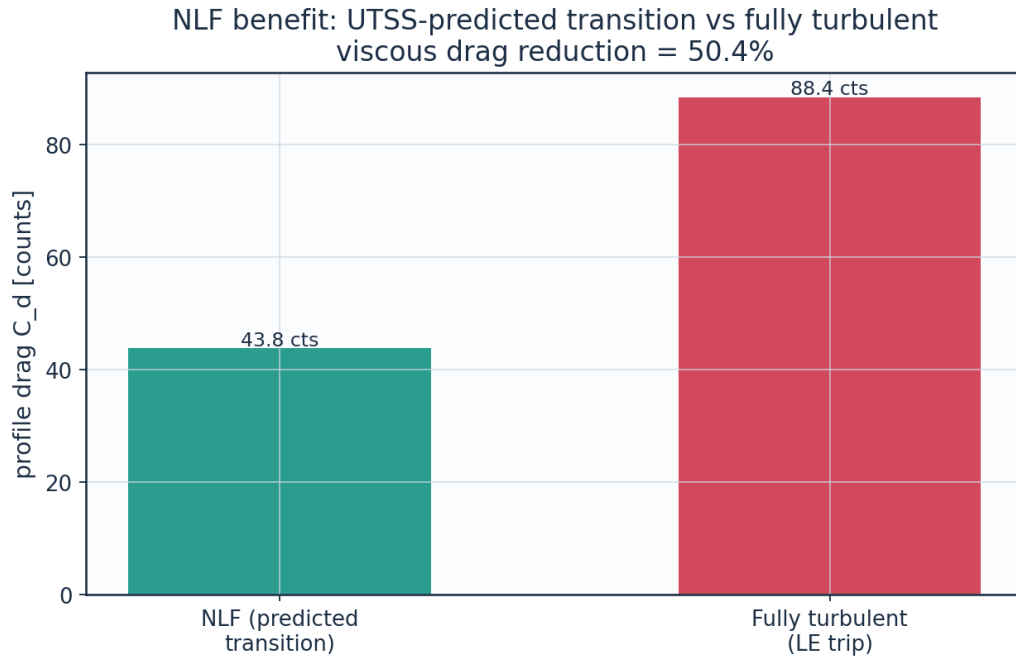


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

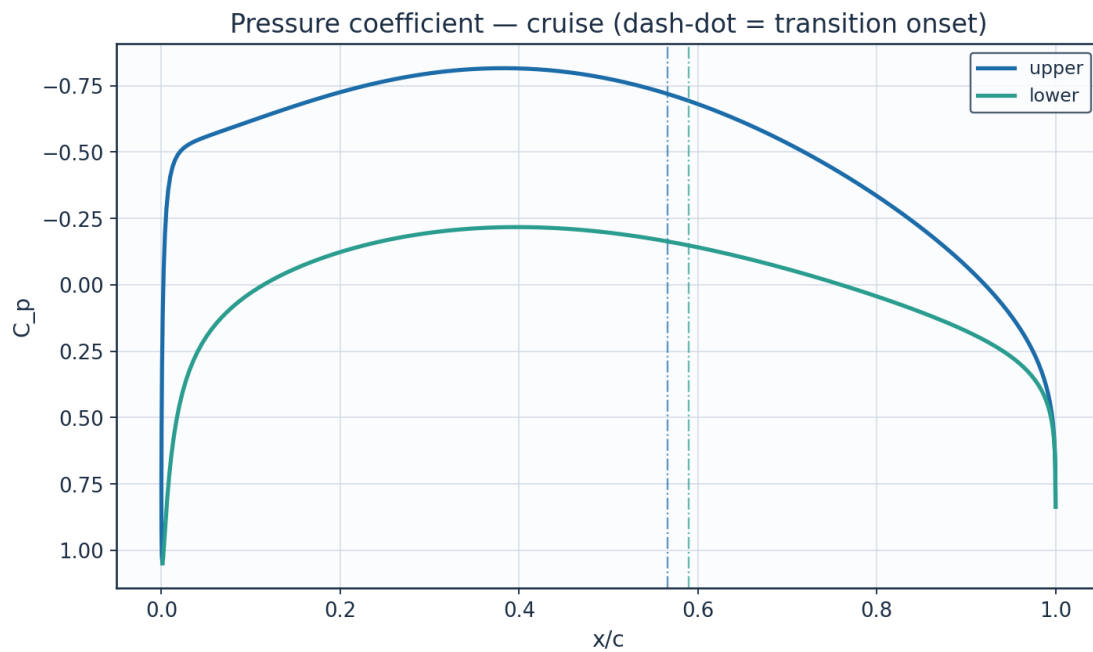


Fig. 13. Pressure coefficient C_p — cruise.

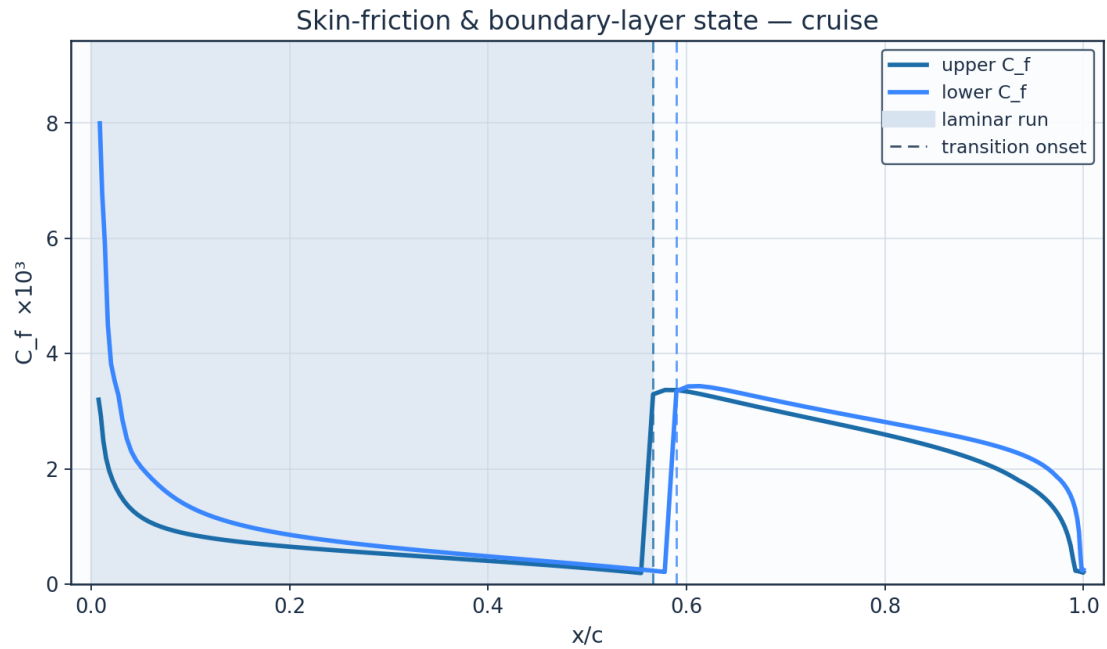


Fig. 14. Skin-friction C_f and laminar run — cruise.

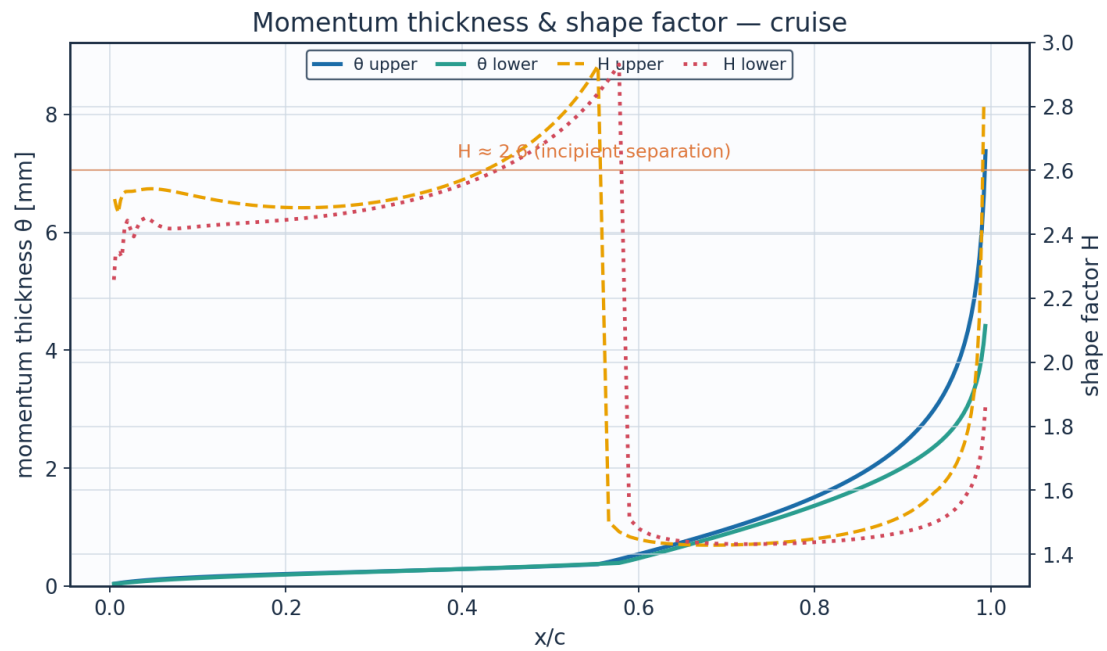


Fig. 15. Momentum thickness θ and shape factor H — cruise.

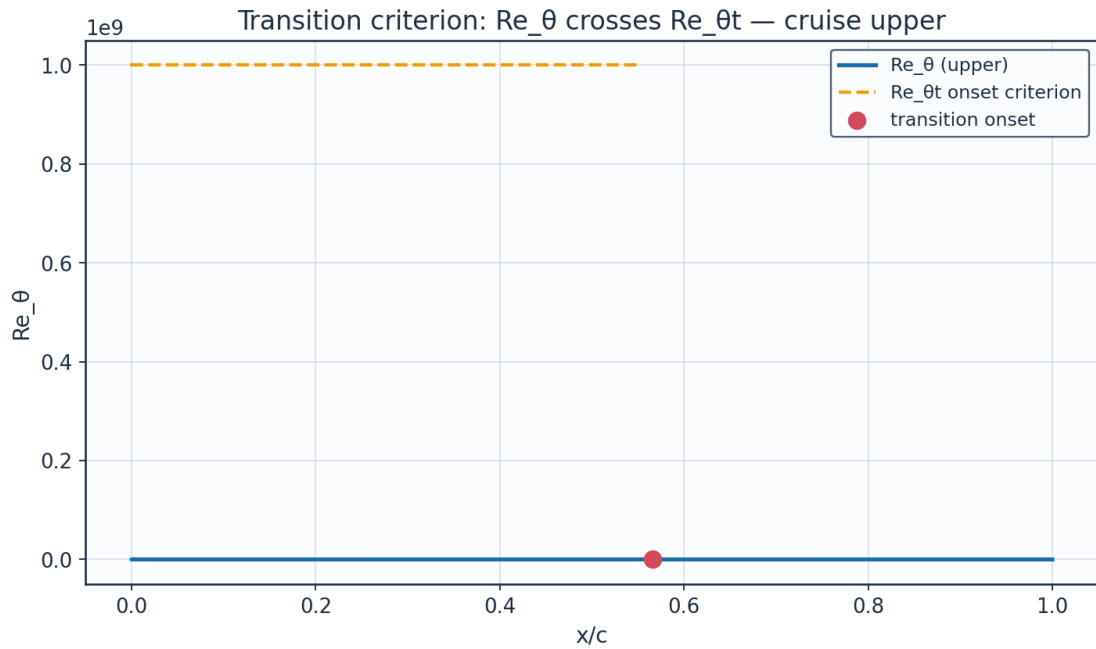


Fig. 16. Transition criterion Re_θ vs $Re_{\theta t}$ — cruise.

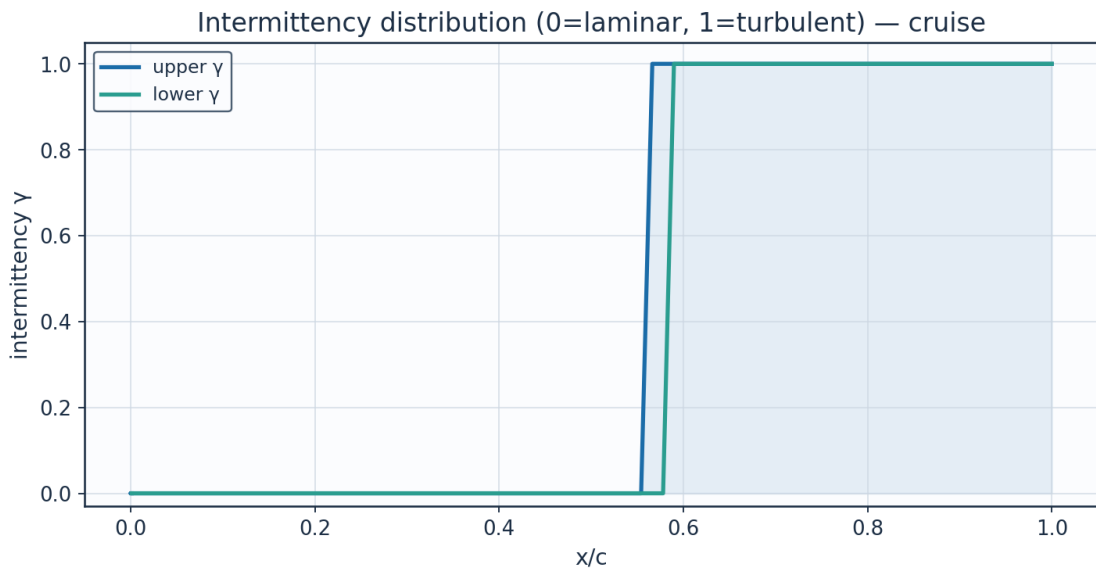


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittyency_gamma	state
0.0011523200325028	0.0	0.0	1.0484500005807595	0.6743779759375519	0.0054416302825443	1e-05	2.6099999414507185	2491.908085201937	0.0001725874393307	10000000.0	0.0	laminar
0.000242727	0.014097598	44712.17719	0.531909959	88.07774049	0.710709004	0.026313920	2.131948169	0.013827901	58.12010330	10000000.0	0.0	laminar

1046013	6799438	636864	2569656	458455	4378861	3079384	0088806	7318326	748996	00.0		r
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1906595115746616	134.13696378207842	1.082365958215552	0.037378612170277	2.462407997730696	0.0042799774898428	122.44703974006676	10000000.0	0.0	laminar
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4431856098056879	146.32725910183538	1.1807307959357134	0.0562800876810639	2.505801090933648	0.0024818494066748	199.367350794746	10000000.0	0.0	laminar
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5161572579089404	149.61652550936128	1.2072722494368673	0.0791442822357182	2.534326561988456	0.0016657296805269	285.952514374348	10000000.0	0.0	laminar
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5481704273505028	151.03025162861496	1.2186797614497848	0.1023649543614522	2.541964874436309	0.0012640951244638	372.9402935748207	10000000.0	0.0	laminar
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5751048472060674	152.20640645138116	1.2281702844634217	0.1246539560144825	2.5379251173937782	0.0010366343638922	457.26533154868616	10000000.0	0.0	laminar
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.6054032741270788	153.51532585711524	1.2387320995430888	0.1455636235289966	2.523781735915112	0.0008978756491207	538.0110117597337	10000000.0	0.0	laminar
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6399761678473315	154.99111243577192	1.250640384249193	0.165128997834491	2.507577796128069	0.0008020183726366	615.4791217278166	10000000.0	0.0	laminar
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6773382556320362	156.56525350268146	1.2633423021710868	0.1835810495681587	2.493799043571883	0.0007282351136715	690.3424030450623	10000000.0	0.0	laminar
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.7150952205439175	158.13484523358554	1.2760075109988878	0.2012557344703509	2.4852163093874604	0.000666060406611	763.4345155900454	10000000.0	0.0	laminar
0.2286307839316745	0.5099588956971477	1617394.0707776232	-0.7504707657473884	159.5866938991024	1.2877226380431313	0.2185549060746073	2.4833880267323125	0.000609960237384	835.6881399908314	10000000.0	0.0	laminar
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7805852261678627	160.80872570370784	1.2975833474841956	0.2359240306060316	2.4896858991225392	0.0005566024948537	908.107089093837	10000000.0	0.0	laminar
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.80268469982041	161.69756919660222	1.3047555236820498	0.2538395197093098	2.5059865978013764	0.0005038714709239	981.752635784148	10000000.0	0.0	laminar

			01									
0.3609 37646 56271 2	0.7791 55626 03463 46	24711 82.873 46131 45	- 0.8143 64452 32521 26	162.16 46593 46477 78	1.3085 24525 62892 9	0.2728 10213 57247 47	2.5344 42538 85766 63	0.0004 50237 07587 32	1057.7 65797 07977 38	1000 0000 00.0	0.0	lam ina r
0.4083 55390 17306 84	0.8751 39546 45295 7	27756 07.063 36620 6	- 0.8137 74739 53945 51	162.14 11197 75158 5	1.3083 34582 23116 16	0.2933 83127 65978 55	2.5788 94097 43799 3	0.0003 94359 20170 89	1137.3 89965 61953 74	1000 0000 00.0	0.0	lam ina r
0.4566 86447 36747 44	0.9728 85870 56001 44	30856 20.921 96654 44	- 0.7997 78064 70728 27	161.58 10430 77768 75	1.3038 15261 57448 34	0.3161 67924 20838 32	2.6468 53101 53594	0.0003 34653 91388 78	1222.0 50272 45808 6	1000 0000 00.0	0.0	lam ina r
0.5054 58667 99852 56	1.0715 44176 36863 5	33985 27.236 81889 24	- 0.7720 27910 22110 03	160.46 27537 07129 7	1.2947 91661 27754 76	0.3418 85299 29879 08	2.7510 96743 41320 56	0.0002 69014 03125 61	1313.5 07267 73401 83	1000 0000 00.0	0.0	lam ina r
0.5541 90867 66523 94	1.1702 45063 74149 47	37115 68.604 06482 8	- 0.7309 53208 29388 42	158.78 78833 16317 2	1.2812 76947 32902 13	0.3714 71687 76999 43	2.9274 15498 98563 2	0.0001 92931 93142 54	1410.3 60191 61985 66	1000 0000 00.0	0.0	tra nsit ion al
0.6023 99717 62090 82	1.2681 03309 91279 1	40219 37.436 53568 5	- 0.6776 53007 07654 81	156.57 84251 17026 46	1.2634 48585 38027 68	0.5502 73601 30119 76	1.4446 86282 67579 36	0.0033 35021 87758 95	2063.9 45426 77916 57		1.0	tur bul ent
0.6496 07126 44050 3	1.3642 23169 95281 22	43267 92.774 79366 8	- 0.6137 23995 79041 99	153.87 22197 38402 4	1.2416 11915 64231 29	0.7439 15535 14569 35	1.4301 09360 00738 2	0.0031 59794 15876 87	2748.1 34247 77703 7		1.0	tur bul ent
0.6953 47671 73678 39	1.4577 06209 48466 56	46232 85.129 50581 1	- 0.5410 53727 68791 76	150.71 74710 46898 48	1.2161 55900 43120 1	0.9453 95204 09354 88	1.4283 90462 64348 42	0.0029 85675 39987 62	3429.5 56788 43587 43		1.0	tur bul ent
0.7391 75615 76998 33	1.5476 61543 81474 4	49085 88.956 03955 5	- 0.4616 12518 26452 6	147.16 68703 97202 25	1.1875 05712 10061 74	1.1589 75728 57876 52	1.4326 63730 85147 42	0.0028 24135 35785 41	4116.8 55580 48418 9		1.0	tur bul ent
0.7806 71085 14247 59	1.6332 17905 01870 1	51799 40.926 63225 7	- 0.3772 69065 09069 94	143.27 17466 98238 22	1.1560 75529 27907 52	1.3896 93401 42396 97	1.4417 22299 53111 8	0.0026 69224 67881 59	4820.2 33993 73280 2		1.0	tur bul ent
0.8194 45102 96803 74	1.7135 36656 13103 57	54346 81.206 41017 8	- 0.2896 42823 00443 23	139.07 64000 66341 6	1.1222 22814 49229 63	1.6435 03375 27774 5	1.4557 64225 67886 52	0.0025 14386 75723 52	5551.1 61799 61272 25		1.0	tur bul ent
0.8551 43310 57486 8	1.7878 24762 77398 5	56702 94.594 42127	- 0.1999 93692 84881 82	134.61 24371 09550 54	1.0862 02604 94725 66	1.9281 29081 06514 12	1.4758 93993 23221 08	0.0023 52943 52711 38	6324.0 90035 25837 1		1.0	tur bul ent

0.8874 48376 11318 26	1.8553 46811 45599 4	58844 48.641 06943 6	- 0.1091 38200 00321 98	129.89 24515 20186 65	1.0481 16520 53658 5	2.2545 45746 25368 04	1.5044 08045 22881 6	0.0021 76913 65081 73	7159.2 96266 55000 6		1.0	tur bul ent
0.9160 81229 91338 4	1.9154 35385 45516 16	60750 26.556 43811 1	- 0.0173 69522 59207 68	124.90 15687 55567 9	1.0078 44536 93447 95	2.6397 45669 75076 06	1.5458 50577 91549 07	0.0019 74912 68758 95	8087.9 36706 77784 8		1.0	tur bul ent
0.9408 01367 76385 24	1.9674 99407 22596 13	62401 53.669 20543 7	0.0756 60067 35794 82	119.58 35624 55574 66	0.9649 32957 43846 64	3.1125 08952 79090 4	1.6045 02549 13778 38	0.0017 42868 63362 19	9162.3 40677 68806		1.0	tur bul ent
0.9614 06511 28321 76	2.0110 30361 48001 73	63782 17.163 87019 8	0.1711 81141 30624 9	113.81 34551 80805	0.9183 73325 30299 52	3.7256 47881 37649 26	1.6856 79107 69095 6	0.0014 81264 89857 48	10475. 95943 58363 8		1.0	tur bul ent

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Ui_nf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011 52320 03250 28	0.0	0.0	1.0484 50000 58075 95	0.6743 77975 93755 19	0.0054 41630 28254 43	1e-05	2.6099 99969 21188 65	2491.9 07990 24486 43	0.0001 72587 43933 07	1000 0000 00.0	0.0	la mi na r
0.0067 20774 97236 91	0.0175 58980 96768 28	55690. 35452 34006 45	0.7967 88059 19874 23	61.869 95831 42874	0.4992 35518 88645 79	0.0406 30933 80364 24	2.3372 93310 87696 8	0.0096 89900 24599 78	63.690 16470 58789 6	1000 0000 00.0	0.0	la mi na r
0.0168 97481 50526 17	0.0419 27102 59539 91	13297 6.6922 10909	0.5091 25482 40715 59	89.950 65248 46617 9	0.7258 21737 89804 35	0.0547 33393 18934 9	2.4177 31536 21650 6	0.0045 05687 02107 84	123.35 42264 46452 88	1000 0000 00.0	0.0	la mi na r
0.0315 87190 34084 9	0.0742 42760 85277 24	23546 9.5685 52791 4	0.3201 16148 47371 03	104.06 54470 09191 12	0.8397 15572 00463 43	0.0748 29513 88036 95	2.4226 59161 11675 76	0.0028 50486 97101 84	193.72 46733 00605 45	1000 0000 00.0	0.0	la mi na r
0.0506 50104 83468 89	0.1147 42805 66774 19	36392 0.1806 98816 6	0.1932 57759 39698 61	112.43 01677 32619 13	0.9072 11426 28479 7	0.0961 96452 33136 3	2.4361 97277 9606 59	0.0020 25747 57906 59	267.79 94049 65490 74	1000 0000 00.0	0.0	la mi na r
0.0739 02667 92458 39	0.1632 86856 80577 34	51788 3.2963 74761 5	0.1005 40820 64840 53	118.11 26263 41393 7	0.9530 63811 66530 09	0.1171 33918 30613 91	2.4177 20290 52703 7	0.0016 27966 74424 97	341.40 97509 38825 4	1000 0000 00.0	0.0	la mi na r
0.1011 19142 10955 74	0.2195 30525 04901 36	69626 6.6450 39893 19	0.0275 01103 73430 19	122.37 11244 77975 4	0.9874 26102 91000 72	0.1379 82018 43587 99	2.4244 69996 95909 5	0.0013 25610 04567 26	415.57 61314 58466 3	1000 0000 00.0	0.0	la mi na r
0.1320 34052 45495 44	0.2829 94629 23130 39	89755 0.4477 80308 7	- 0.0329 49480 64718 06	125.76 58705 52816 2	1.0148 18683 48282 34	0.1583 16414 18445 45	2.4315 11001 61082 74	0.0011 16263 92501 32	488.98 45088 28673 1	1000 0000 00.0	0.0	la mi na r
0.1663	0.3531	11199	-	128.55	1.0373	0.1781	2.4380	0.0009	561.44	1000	0.0	la

45511 63160 67	01330 53650 2	02.021 44727 8	0.0840 89705 79544 8	35542 38377 76	12810 66783	59494 28431 66	06590 24371 67	63948 87007 87	44888 69421 5	0000 00.0		mi na r
0.2037 19371 58191 91	0.4291 98466 63705 09	13612 52.957 21369 96	- 0.1271 71832 79815 87	130.84 60216 58987 44	1.0558 10983 18064 53	0.1976 99228 27537 9	2.4459 91443 30303 4	0.0008 45780 76665 47	633.16 23487 04087 3	1000 0000 00.0	0.0	la mi na r
0.2437 94056 30589 38	0.5105 78867 09950 84	16193 60.381 63382 23	- 0.1623 62939 49216 68	132.68 26339 36490 7	1.0706 30810 25560 3	0.2172 10641 51975 21	2.4575 27586 34951 5	0.0007 48679 36555 84	704.53 83141 18482 3	1000 0000 00.0	0.0	la mi na r
0.2861 85839 50534 81	0.5964 96549 11452 82	18918 58.323 28838 8	- 0.1893 00550 93779 56	134.06 75651 33079 26	1.0818 05972 86099 4	0.2370 30658 43315 61	2.4743 49612 42659 05	0.0006 64698 12331 22	776.11 42268 61983 8	1000 0000 00.0	0.0	la mi na r
0.3304 94258 79381 13	0.6861 79937 75148 77	21762 99.642 36291 36	- 0.2074 61412 92644 59	134.99 13376 06891 08	1.0892 59994 85909 8	0.2575 36478 17414 93	2.4985 14016 58206 2	0.0005 88804 34298 5	848.52 56312 97554 5	1000 0000 00.0	0.0	la mi na r
0.3763 07324 00294 27	0.7788 41767 61081 37	24701 87.434 89202	- 0.2164 23488 27837 99	135.44 43196 35477	1.0929 15156 81893 9	0.2791 33956 63293 01	2.5325 38778 11893 7	0.0005 17586 54312 33	922.48 06156 02088 4	1000 0000 00.0	0.0	la mi na r
0.4232 06194 57674 89	0.8736 85455 08345 47	27709 95.243 12299 02	- 0.2160 46718 78413 36	135.42 53140 60304 83	1.0927 61798 73625 5	0.3022 48373 59043 62	2.5797 25554 16770 53	0.0004 48591 88123 06	998.74 20663 0817	1000 0000 00.0	0.0	la mi na r
0.4707 69074 33519 84	0.9699 08166 57242 24	30761 76.786 73797 05	- 0.2065 75740 70034 36	134.94 64691 14026 26	1.0888 97946 04045 75	0.3273 20763 27692 84	2.6461 14850 38042 3	0.0003 79712 93225 74	1078.1 24445 41371 02	1000 0000 00.0	0.0	la mi na r
0.5185 74193 40729 35	1.0667 01325 94151 12	33831 67.572 28672 46	- 0.1886 63903 91346 69	134.03 50378 60133	1.0815 43506 70731 1	0.3548 16985 49342 18	2.7400 55134 98381 34	0.0003 09010 81294 09	1161.5 28489 63853 17	1000 0000 00.0	0.0	la mi na r
0.5662 01898 02471 88	1.1632 49726 11984 71	36893 82.075 53692 4	- 0.1633 24624 77316 74	132.73 23845 84628 68	1.0710 32253 72372 25	0.3852 57693 37011 68	2.8829 58974 78673 7	0.0002 33369 87397 91	1250.0 37146 77680 7	1000 0000 00.0	0.0	la mi na r
0.6132 36022 35407 45	1.2587 30610 74833 58	39922 10.828 80882 6	- 0.1318 26376 04350 61	131.09 07600 19421 88	1.0577 85804 00955 62	0.5238 53956 03922 41	1.4616 08274 43550 32	0.0034 33677 93708 03	1677.4 36316 97706 68		1.0	tur bu len t
0.6592 64843 41077 2	1.3523 14020 52265 4	42890 21.519 44244 1	- 0.0955 55444 09005 39	129.16 85290 40938 6	1.0422 75110 19127 53	0.7120 47426 07735 53	1.4377 55866 04188 8	0.0032 94358 37394 82	2249.6 30284 72964		1.0	tur bu len t
0.7038 81995	1.4431 65382	45771 67.201	- 0.0558	127.02 45984	1.0249 75497	0.9030 96814	1.4319 07233	0.0031 32193	2810.0 03826		1.0	tur bu

09891 86	46492 15	97440 2	72264 25655 04	3724	64010 8	06001 56	10270 94	26112 75	07746 8			len t
0.7466 87725 12314 54	1.5304 50847 76179 95	48540 03.227 71357 7	- 0.0139 82254 57569 55	124.71 27010 62973 96	1.0063 20542 68780 44	1.0978 53101 05836 1	1.4319 32168 39038 36	0.0029 85786 66746 26	3359.0 58916 57851 2		1.0	tur bu len t
0.7872 90817 05456 05	1.6133 45372 85086 3	51169 12.874 84830 4	0.0291 67159 17160 43	122.27 59555 95345 72	0.9866 58174 69741 42	1.2977 49111 15486 91	1.4354 63374 44317 73	0.0028 53014 35098 83	3899.3 85795 16750 5		1.0	tur bu len t
0.8253 11381 55209 68	1.6910 43098 74663 05	53633 40.267 68853 3	0.0729 62068 85514 11	119.74 17781 22673 58	0.9662 09617 11001 26	1.5043 06750 37682 06	1.4418 95935 80190 43	0.0027 28961 33931 73	4433.6 58480 59342 9		1.0	tur bu len t
0.8603 84569 42783 87	1.7627 69288 93695 93	55908 28.239 09553 2	0.1171 78365 82090 99	117.11 67909 03076	0.9450 28305 65689 84	1.7197 22409 75426 1	1.4513 49741 88421 45	0.0026 08532 76645 61	4965.7 39145 61021 8		1.0	tur bu len t
0.8921 65104 53399 4	1.8277 92963 07022 2	57970 58.399 69313 5	0.1620 35480 87852 07	114.38 07835 11247 76	0.9229 51160 18674 96	1.9479 12089 08368 32	1.4645 77134 99257 05	0.0024 85874 62628 97	5502.6 17926 28360 85		1.0	tur bu len t
0.9203 32404 83917 28	1.8854 39395 75766 85	59798 90.779 27608 8	0.2082 75008 29401 14	111.47 77474 37842 15	0.8995 26241 85901 14	2.1963 14608 68757 2	1.4832 65187 97923 76	0.0023 53020 44221 59	6057.5 50595 21170 9		1.0	tur bu len t
0.9445 95976 40102 88	1.9351 01792 46614 69	61374 00.858 26456 7	0.2573 13910 01485 05	108.29 96644 45970 88	0.8738 81939 60408 83	2.4793 88530 77619 8	1.5109 43442 07572 9	0.0021 97337 12897 75	6655.8 62253 39916 4		1.0	tur bu len t
0.9647 00736 13354 88	1.9762 51652 32757 04	62679 12.434 56211 6	0.3115 83506 76964 56	104.65 22734 79347 6	0.8444 50739 52869 62	2.8265 81132 52818 67	1.5556 74939 64634 85	0.0019 95530 12866 36	7347.7 39728 56981 8		1.0	tur bu len t
0.9804 31941 29791 1	2.0084 47529 88663 1	63700 25.412 45964 3	0.3753 69091 58485 07	100.17 04201 74899 48	0.8082 86170 79481 93	3.3046 61215 61795 7	1.6273 60547 73997 26	0.0017 30107 23121 49	8243.0 54819 78839 6		1.0	tur bu len t

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated T_u)

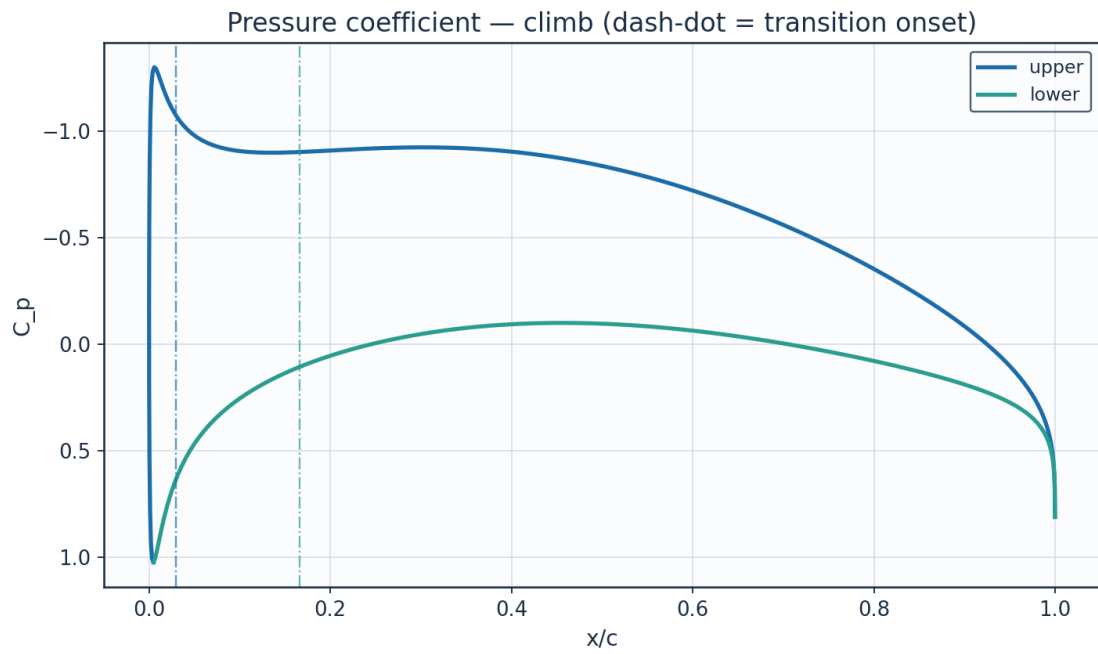


Fig. 18. Pressure coefficient C_p — climb.

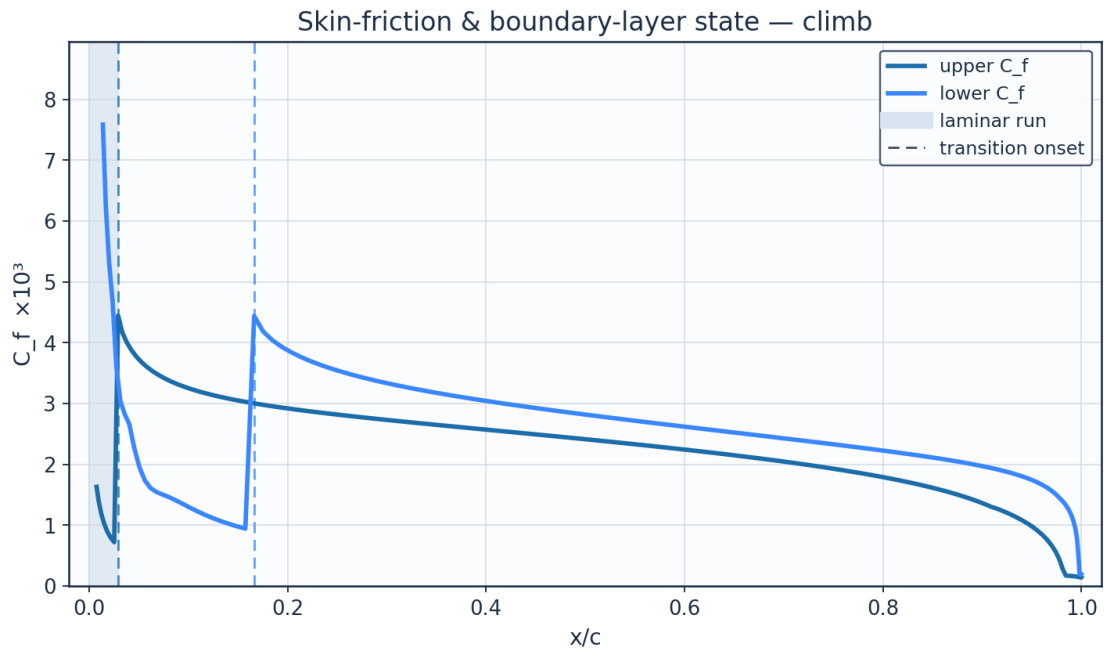


Fig. 19. Skin-friction C_f and laminar run — climb.

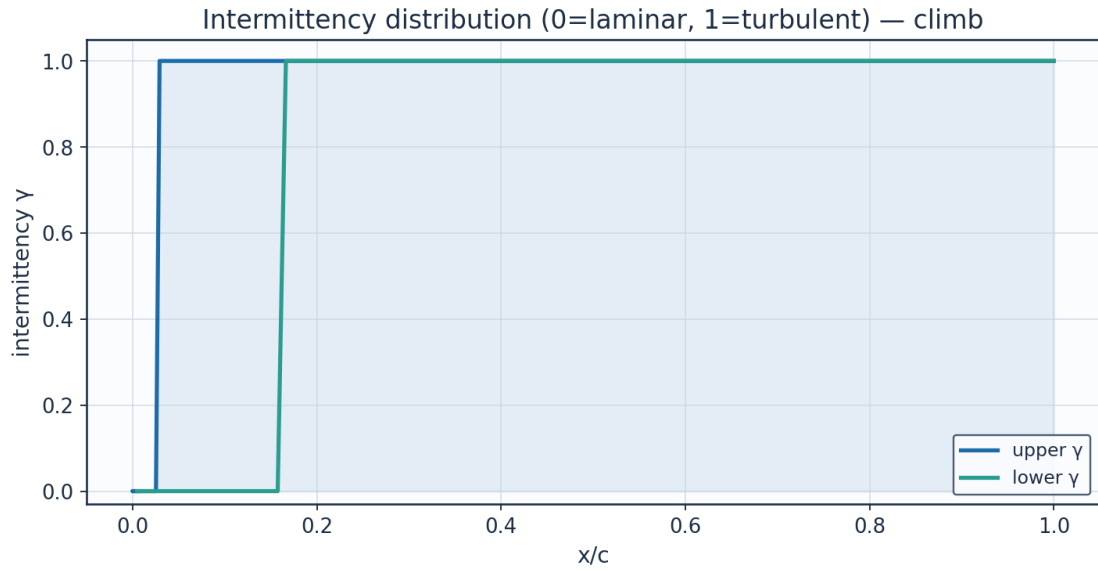


Fig. 20. Intermittency γ — climb.

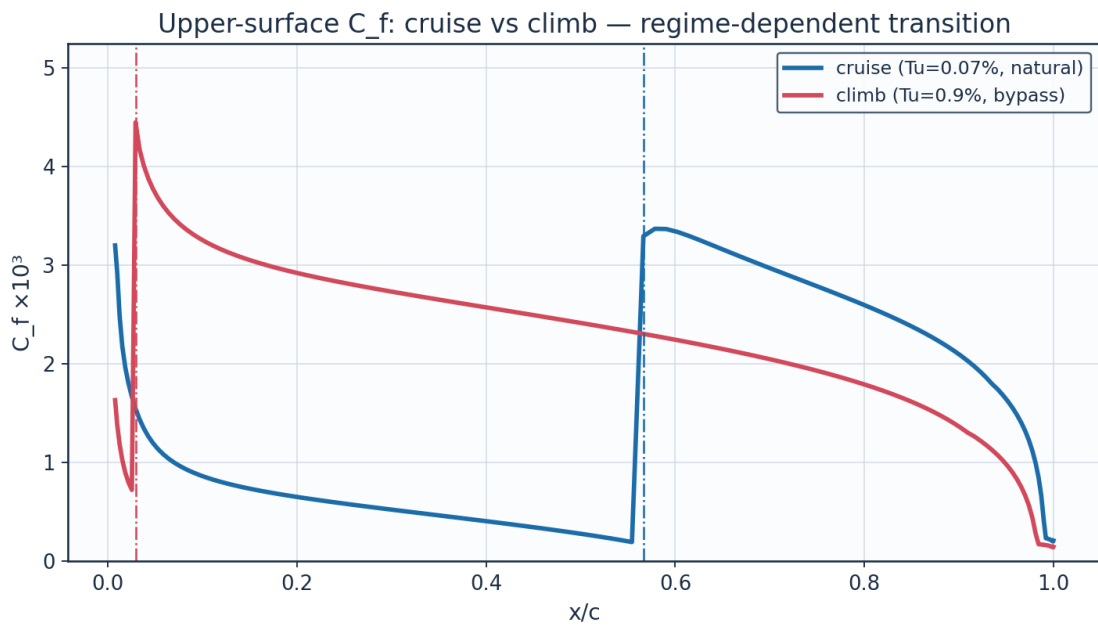


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1114	0.00568	-19.6	0.673	0.149
-2.0	0.0287	0.00456	6.3	0.65	0.4
-1.0	0.1683	0.00435	38.7	0.626	0.471
0.0	0.3077	0.00425	72.5	0.602	0.531
1.0	0.4472	0.00434	103.1	0.566	0.566
2.0	0.5872	0.00447	131.3	0.53	0.602
3.0	0.7281	0.00523	139.1	0.408	0.625

4.0	0.8704	0.00768	113.3	0.048	0.659
5.0	1.0146	0.00836	121.3	0.025	0.682
6.0	1.1612	0.00901	128.8	0.018	0.715
7.0	1.3111	0.00967	135.6	0.022	0.736
8.0	1.4654	0.01061	138.2	0.015	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

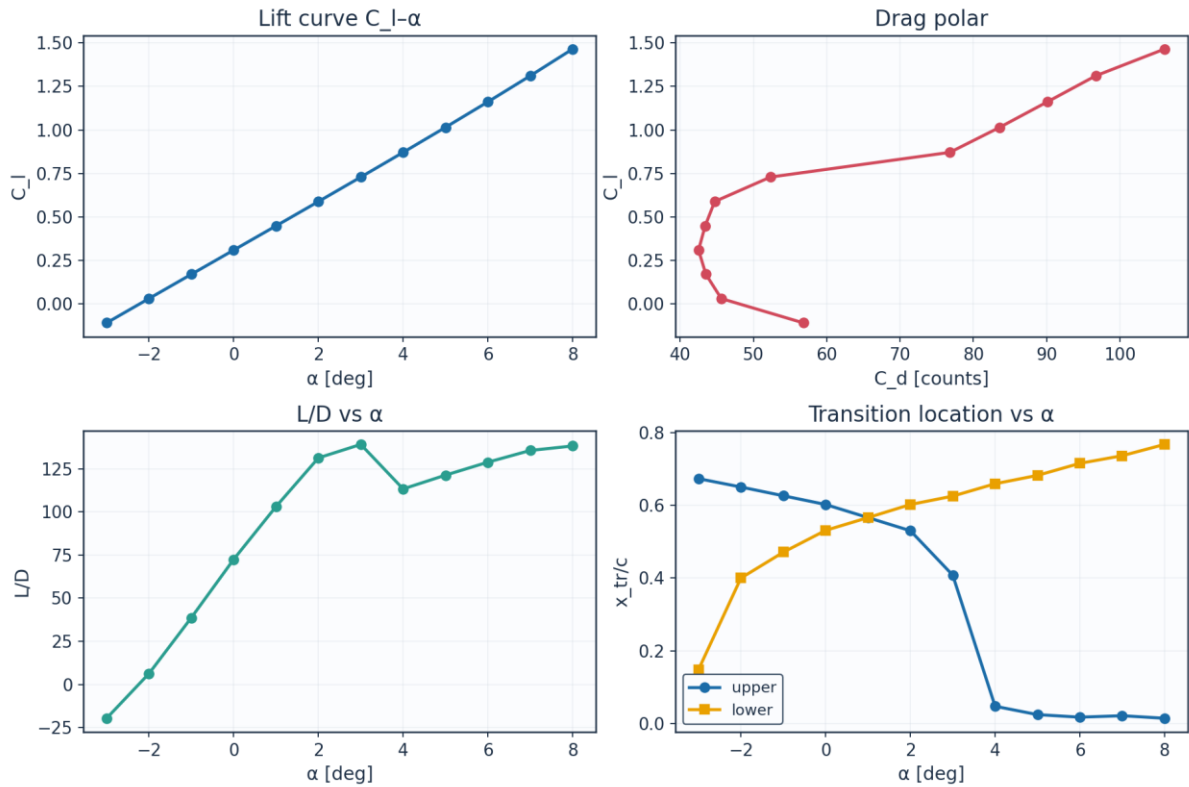


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.53	0.566	0.00431	0.548
0.089	0.757	2.484	7878900.0	1.23	0.542	0.554	0.00431	0.548
0.178	1.515	2.368	7511500.0	0.97	0.554	0.554	0.00428	0.554
0.267	2.272	2.253	7144200.0	0.7	0.566	0.542	0.0043	0.554
0.356	3.029	2.137	6776900.0	0.43	0.578	0.542	0.00428	0.56
0.445	3.786	2.021	6409500.0	0.16	0.59	0.531	0.00431	0.56
0.535	4.544	1.905	6042200.0	-0.1	0.602	0.531	0.00431	0.566
0.624	5.301	1.789	5674900.0	-0.37	0.614	0.519	0.00435	0.566
0.713	6.058	1.673	5307600.0	-0.64	0.626	0.519	0.00436	0.572
0.802	6.815	1.558	4940200.0	-0.91	0.684	0.507	0.0042	0.595
0.891	7.573	1.442	4572900.0	-1.17	0.695	0.495	0.00428	0.595
0.98	8.33	1.326	4205600.0	-1.44	0.707	0.495	0.00433	0.601

Table 18. Span-wise distribution (spanwise_distribution.csv).

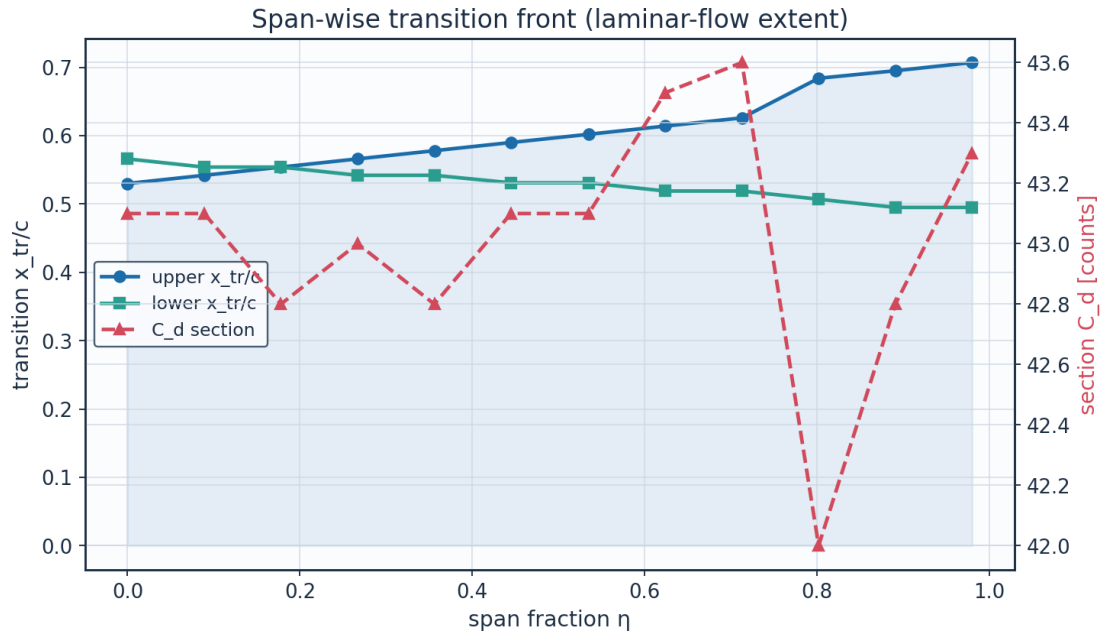


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

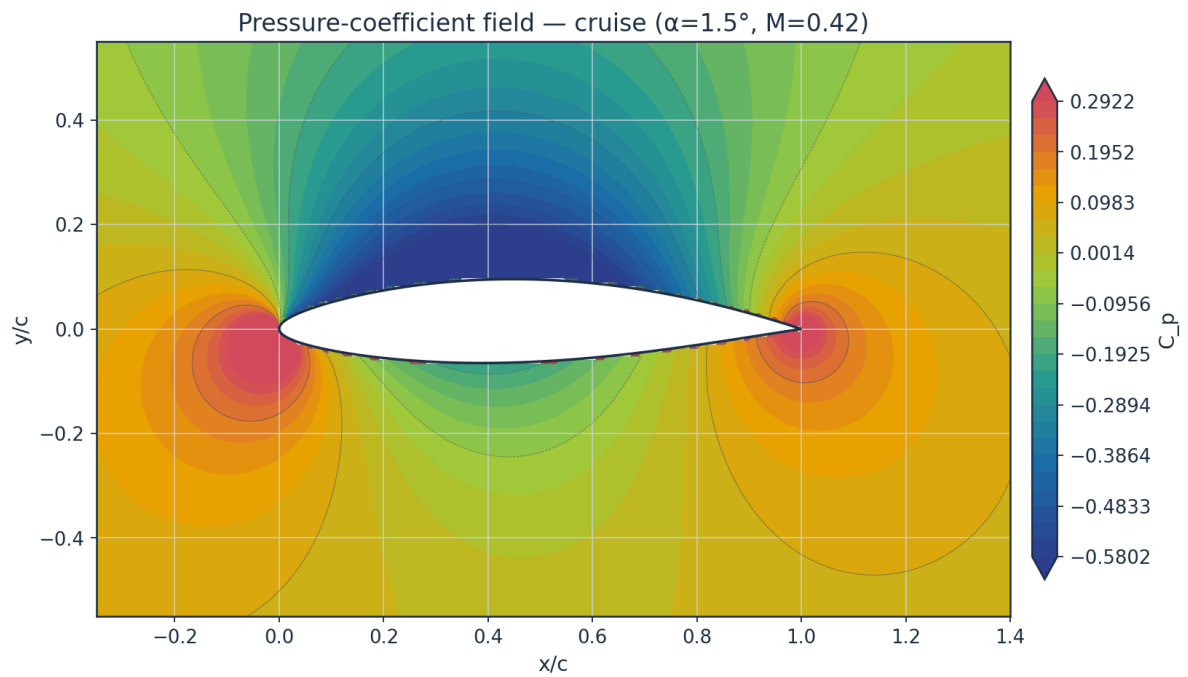


Fig. 24. Pressure-coefficient contour — cruise.

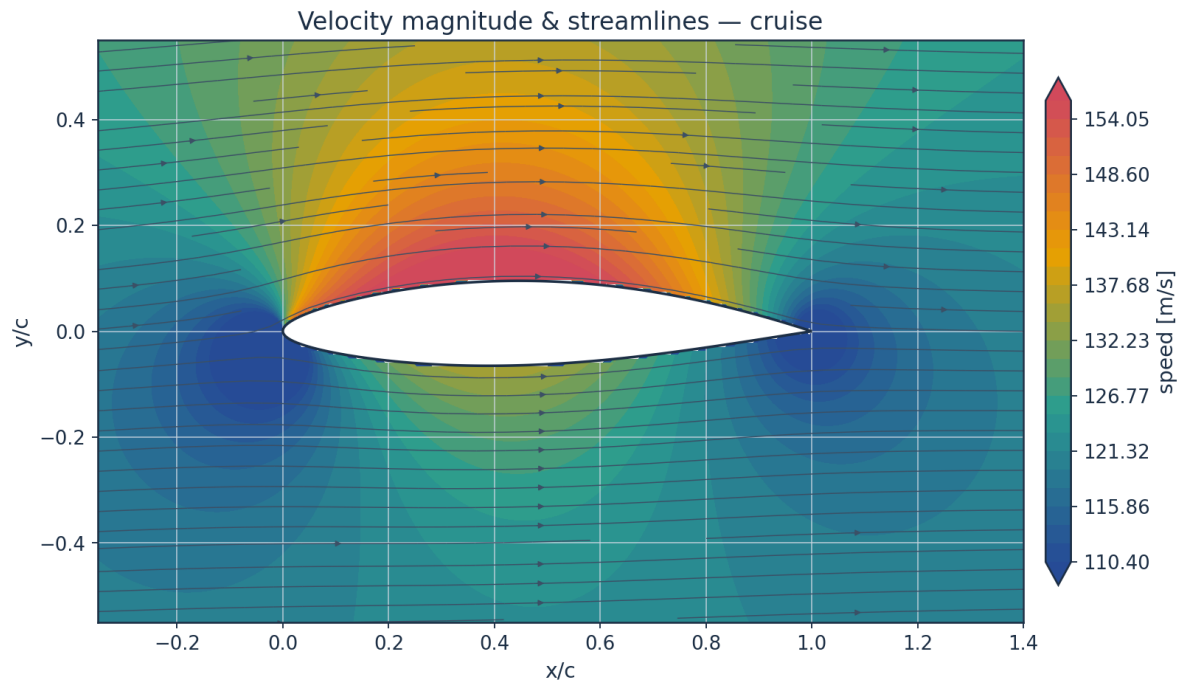


Fig. 25. Velocity magnitude and streamlines — cruise.

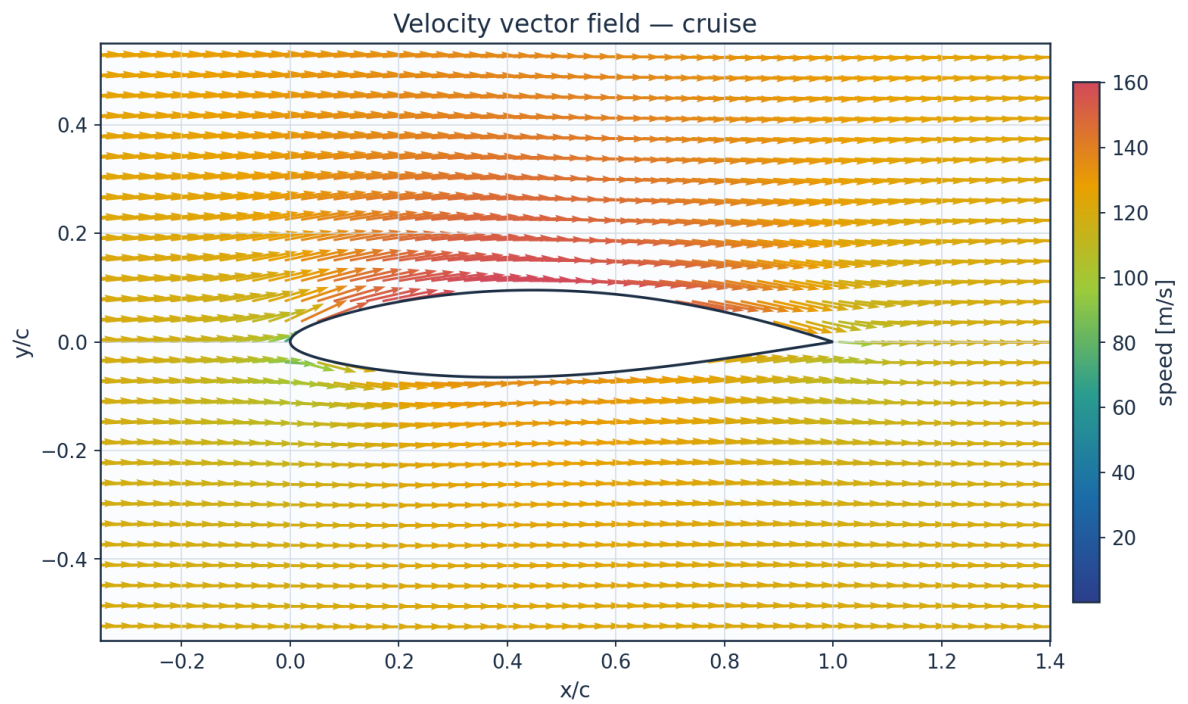


Fig. 26. Velocity vector field — cruise.

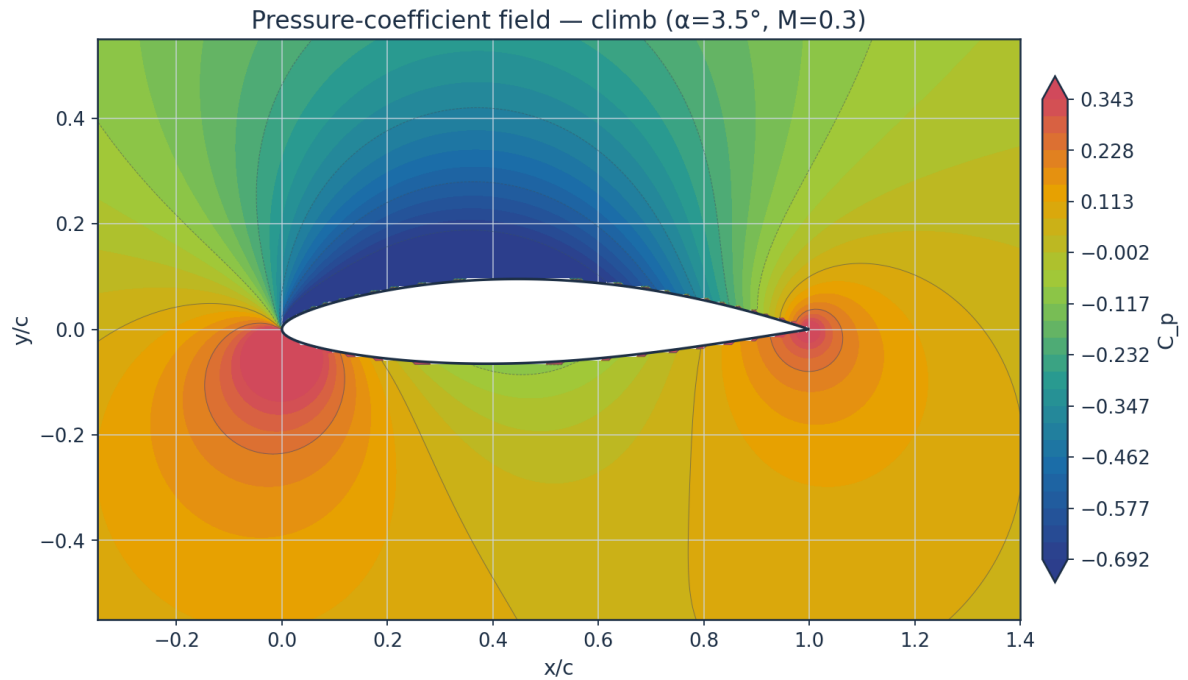


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

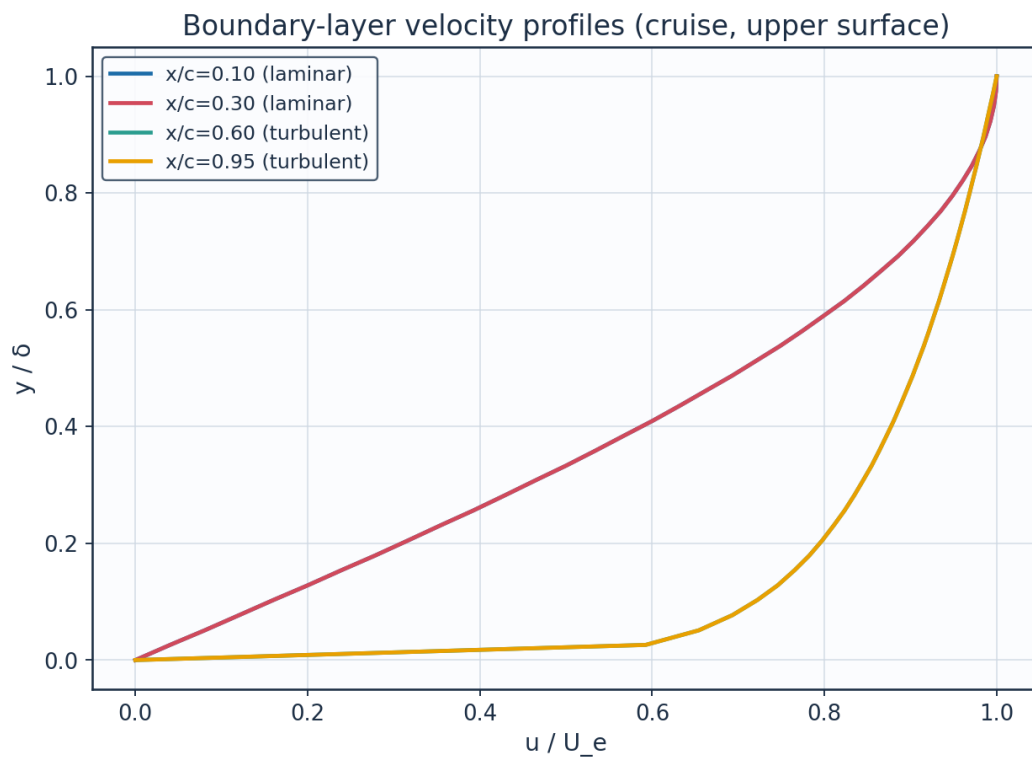


Fig. 28. Boundary-layer velocity profiles.

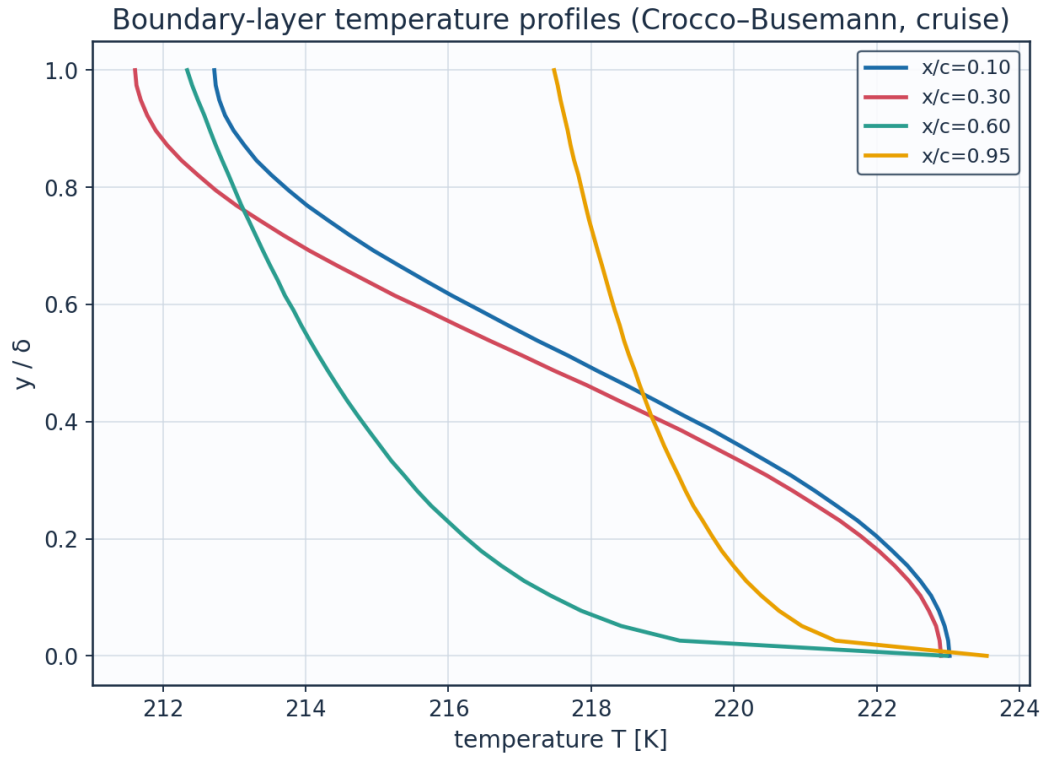


Fig. 29. Boundary-layer temperature profiles (K).

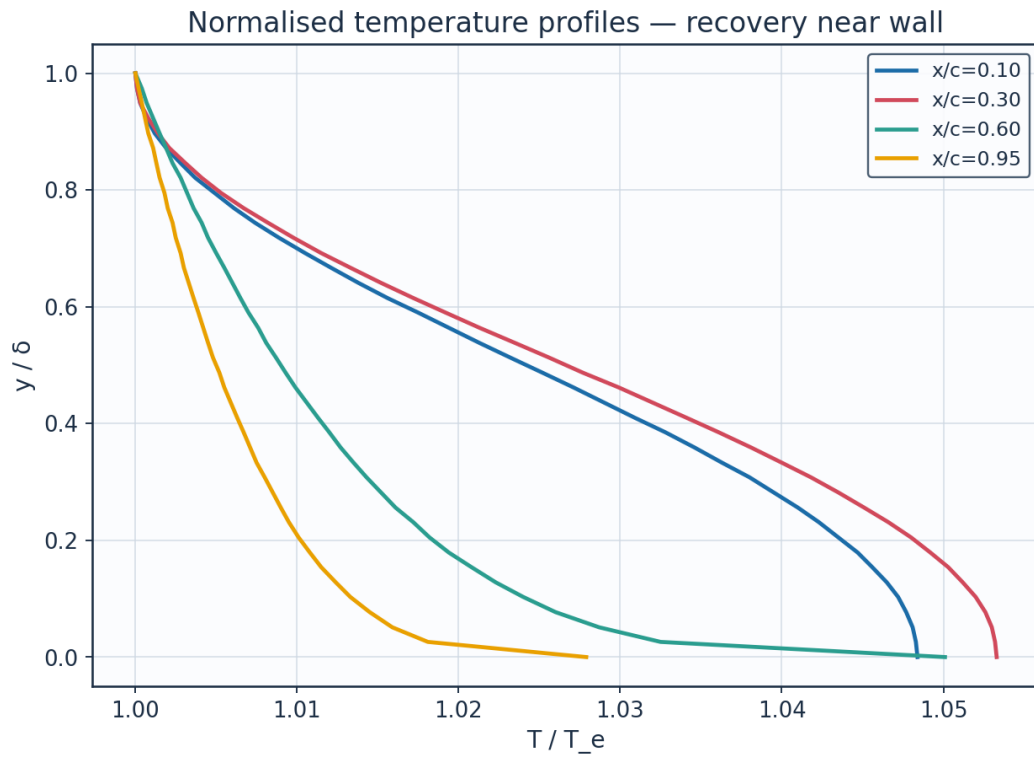


Fig. 30. Normalised temperature profiles T/T_e .

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.2471	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.4942	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.7413	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9884	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2355	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4826	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.7297	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.4091	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.8182	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.2272	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.6363	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	2.0454	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.4545	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.8635	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	0.7901	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	1.5803	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	2.3704	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	3.1605	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	3.9507	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	4.7408	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	5.531	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	4.8772	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	9.7545	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	14.6317	0.8724	1.0067	218.93	0.396	turbulent

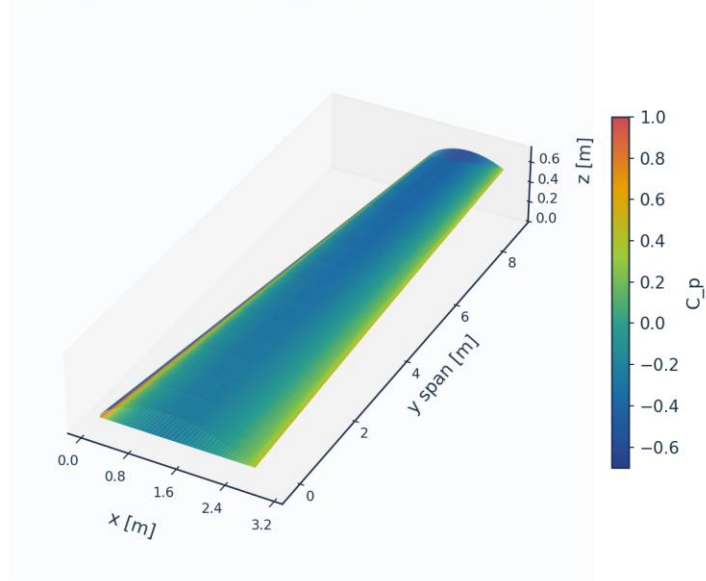
Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

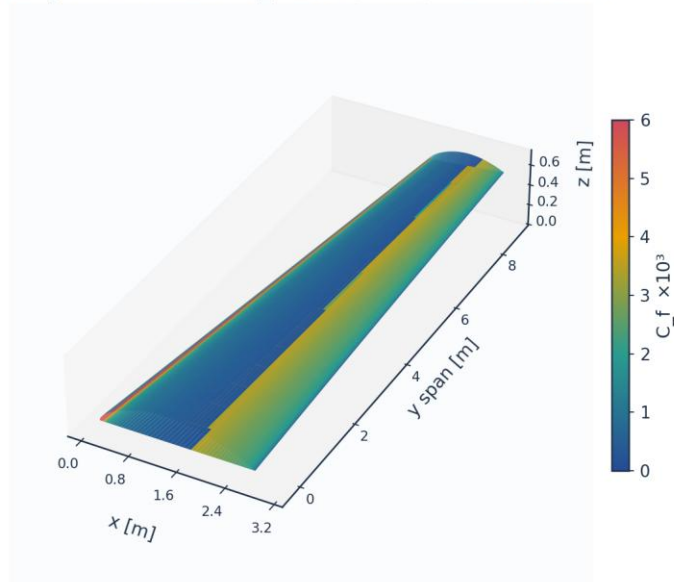
3D wing surface contour: C_p (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 31. 3-D wing surface contour — pressure C_p .

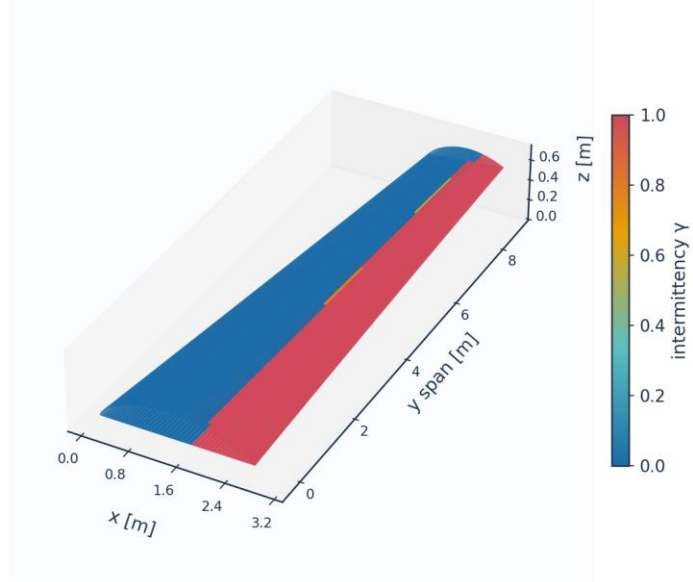
3D wing surface contour: $C_f \times 10^3$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 32. 3-D wing surface contour — skin friction $C_f (\times 10^3)$.

3D wing surface contour: intermittency γ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

Upper-surface flow direction, coloured by C_f (strip formulation: chordwise only)

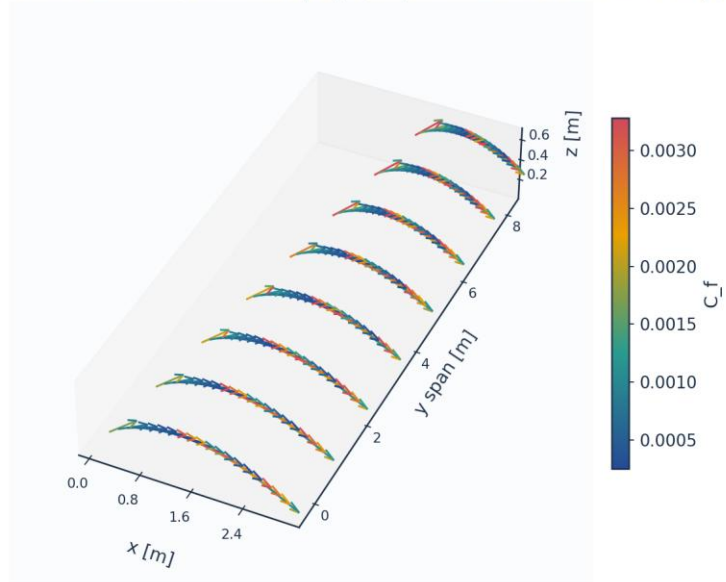


Fig. 34. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

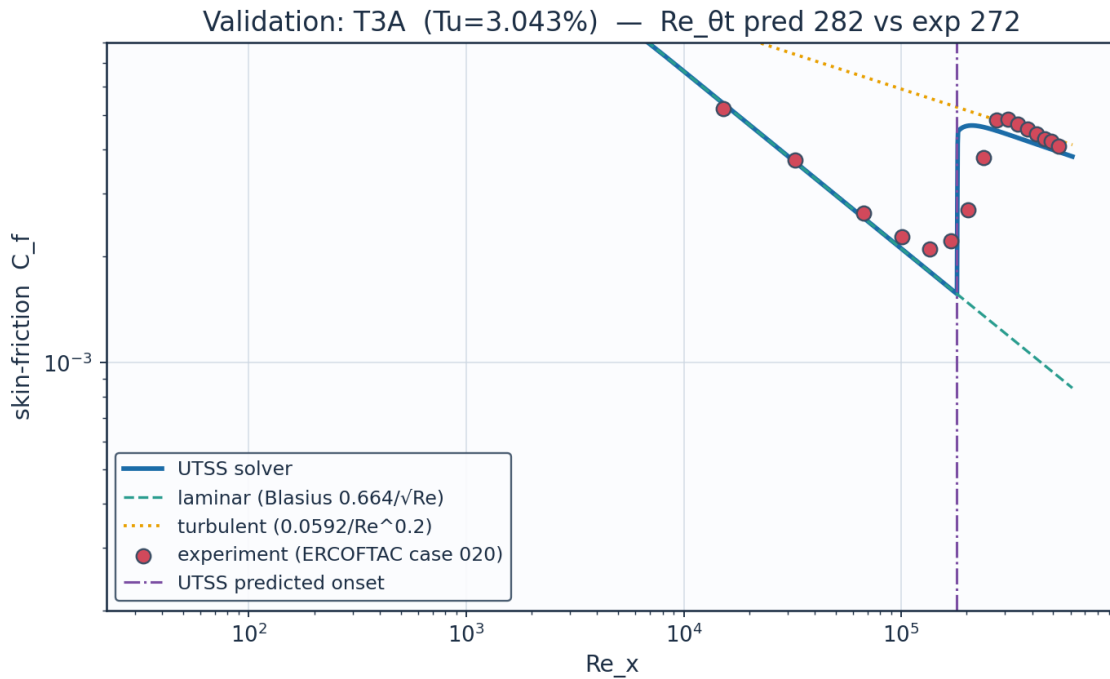
To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ is the metric on the plates and the transition location x_{tr}/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	Cf_err_laminar_pct	n_pts_laminar	Cf_err_turbulent_pct	n_pts_turbulent	Cf_err_all_pts_pct	Re_theta_meas_over_blasius	mechanism
ERCOFT3A flat plate (bypass transition)	3.043	1.529999999999998	272.3	282.1	3.6	134800.0	180400.0	3.8	4	8.2	9	12.7	1.117	bypass
ERCOFT3A flat plate (low-Tu bypass)	0.874	0.98	818.8	683.6	-16.5	144300.0	106000.0	4.2	8		0	216.9	1.027	bypass
ERCOFT3A	5.952	3.83	181.3	168.0	-7.3	59100.0	64050.0	8.2	3	9.7	11	10.1	1.123	bypass

B fla t pla te (hi gh- Tu by pa ss)														
ER CO FT AC T3 C4 fla t pla te (la mi nar se par ati on bu bbl e)	2.1 1		381. 3	271. 4	-28.8	178 500 .0	169 200 .0	21.9	9		0	220.9	1.359	sep ara tio n
Sc hu ba uer & Skr am sta d fla t pla te (na tur al tra nsi tio n)	0.0 3		1100 .0	1161 .6	5.6	280 000 0.0	306 000 0.0		0		0			TS- nat ura l

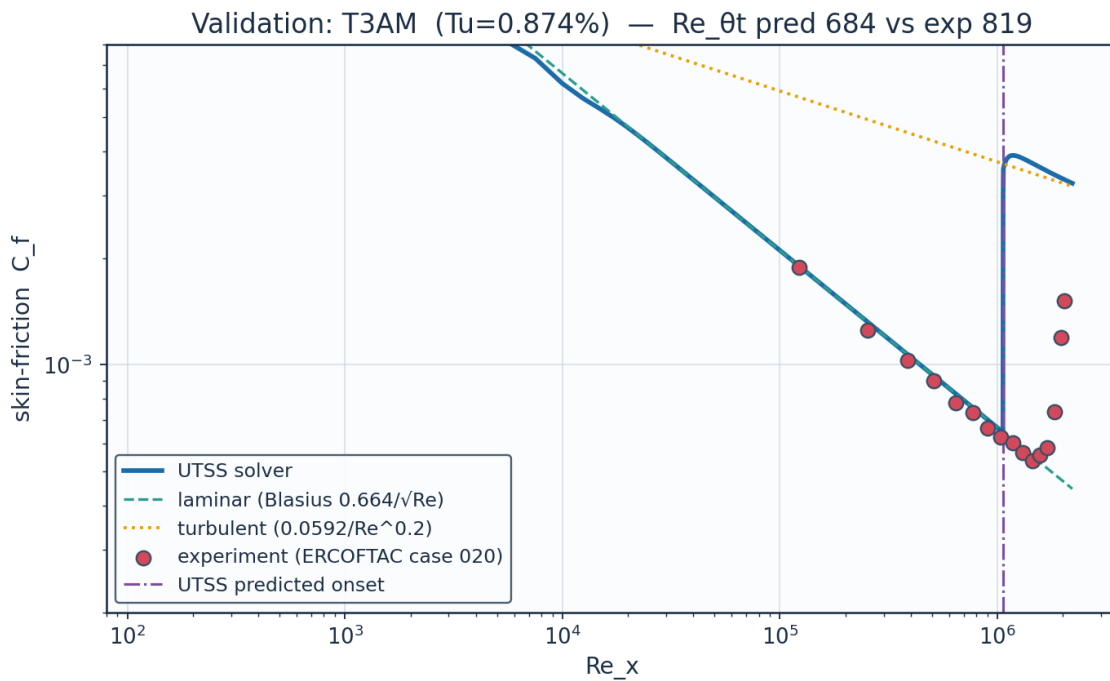
Table 20. Validation summary — predicted vs published $Re_{\theta t}$.

11.1 Flat plates



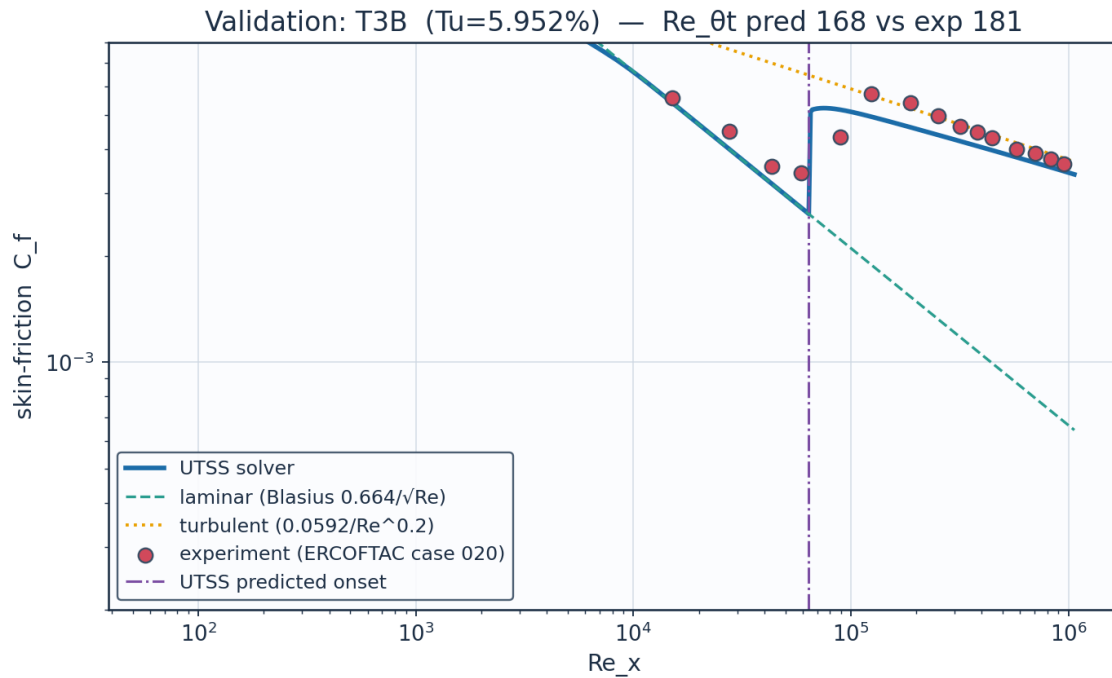
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate ($Tu = 3.0\%$, bypass).



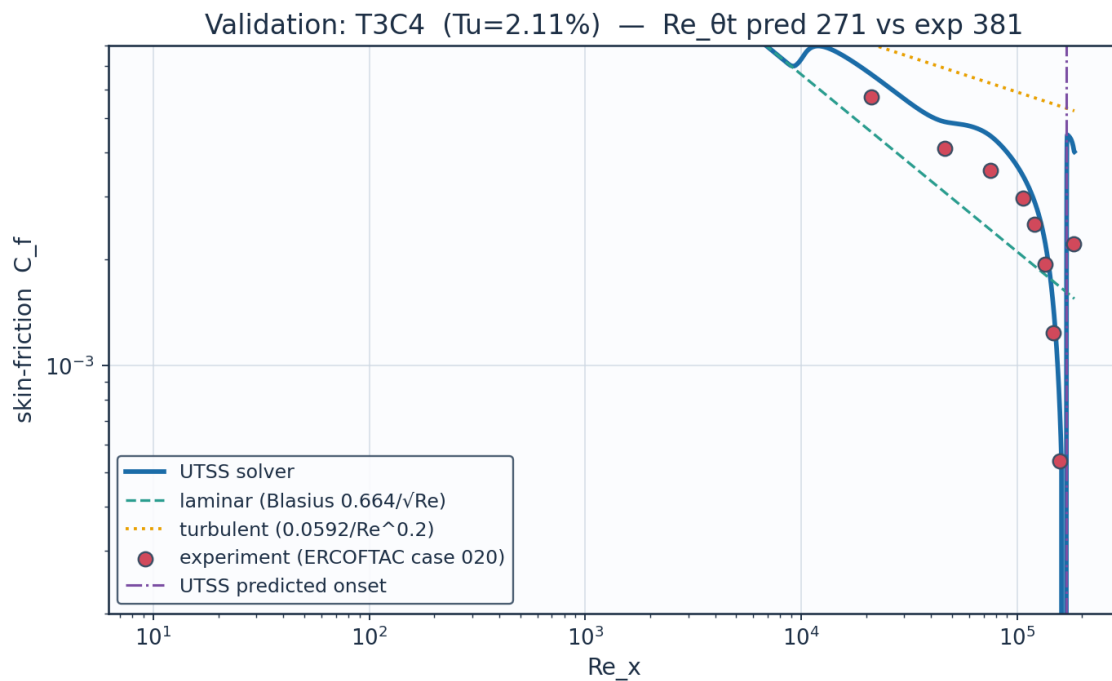
Source: Roach & Brierley (1990), ERCOFTAC T3A-; ERCOFTAC Classic Collection Case 020....

Fig. 36. Validation — ERCOFTAC T3A⁻ flat plate ($Tu = 0.87\%$, bypass).



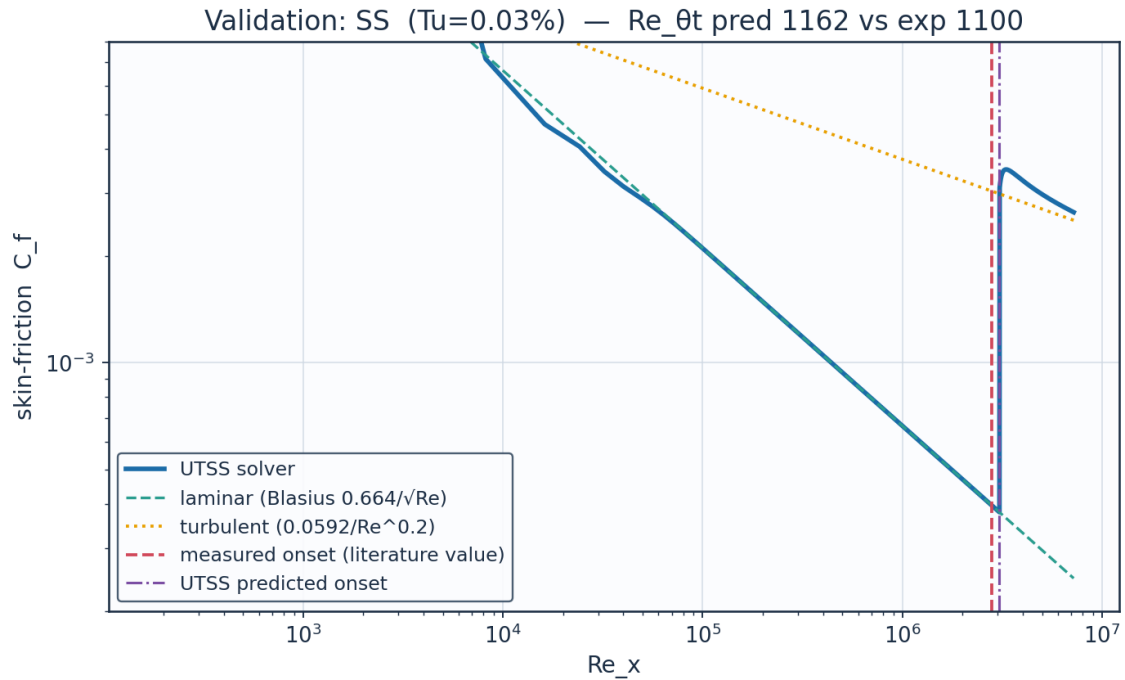
Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 37. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-grad...

Fig. 38. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 39. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

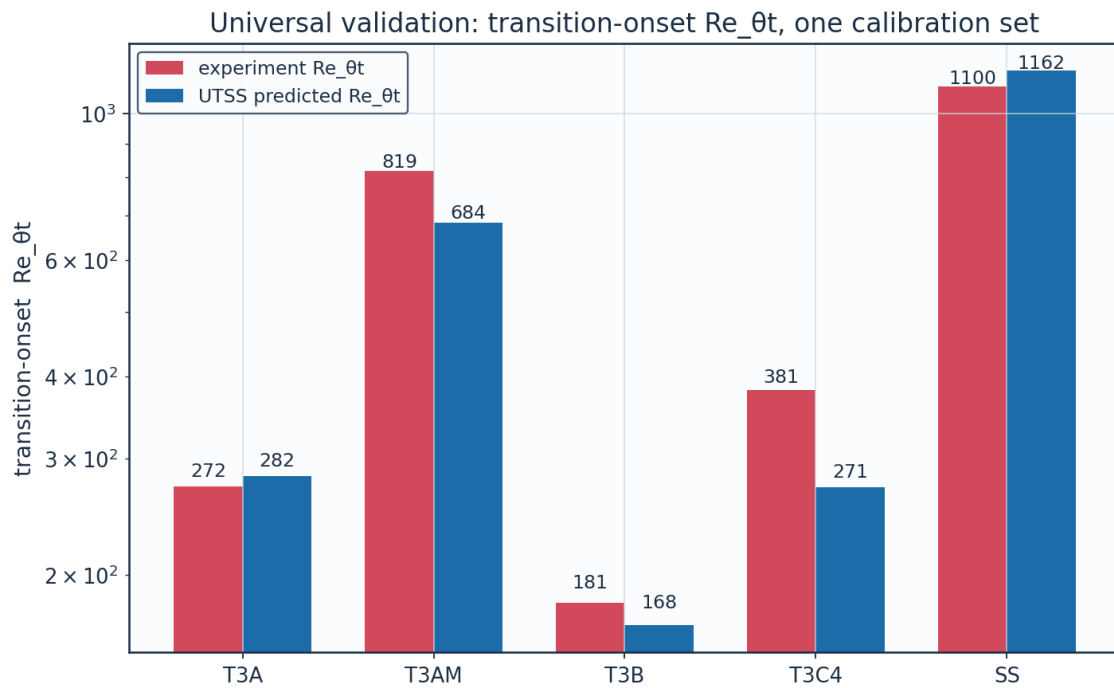


Fig. 40. Universal validation — $Re_{\theta t}$ across all five plates.

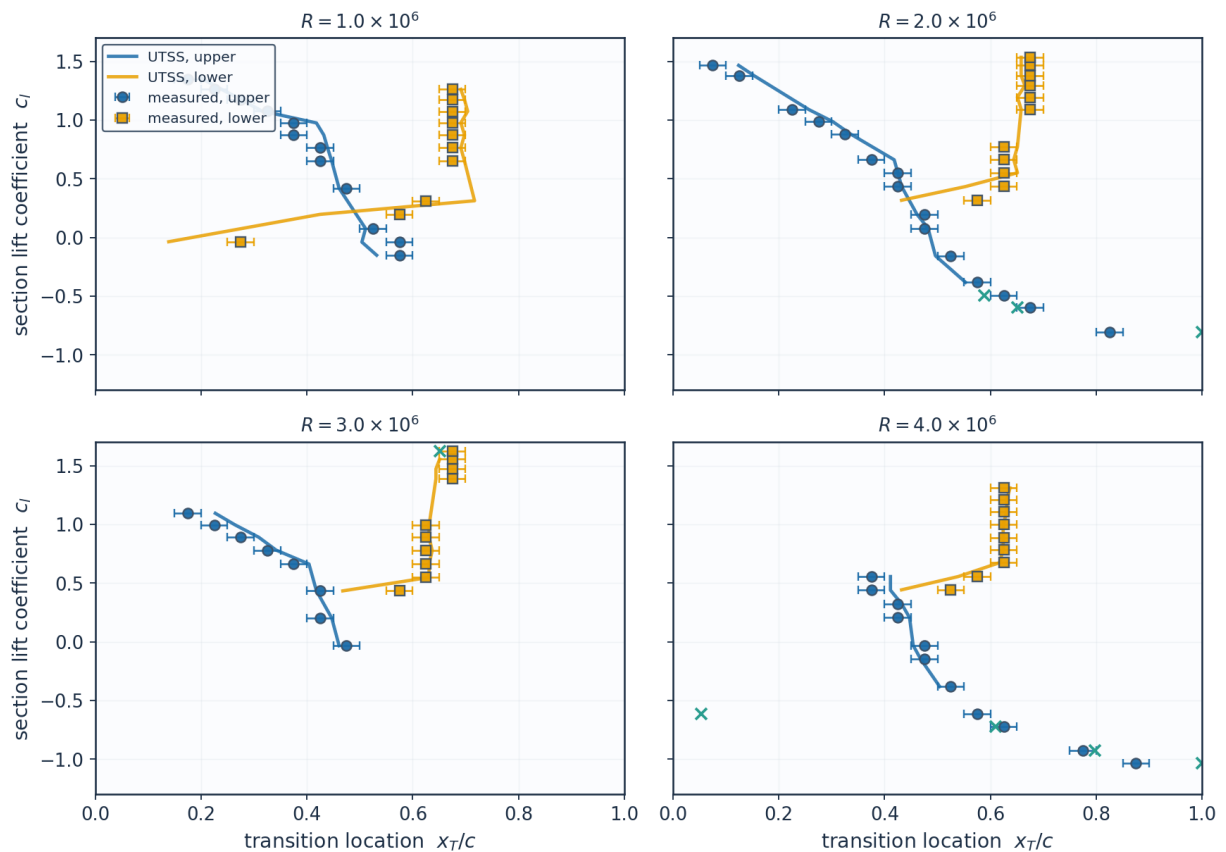
11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and the one on which nothing is calibrated. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is $\pm 0.025c$ and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0409	0.0025	25	54.3
Lower surface	40	0.0332	-0.0148	27	67.5
All	86	0.0373	-0.0055	52	60.5
Conditions the method accepts	83	0.0288	-0.0031	52	62.7

Table 21. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).

NLF(1)-0416 transition location, $M = 0.10$ — 86 points digitised from NASA TP-1861, Fig. 9



Bars show the $\pm 0.025c$ orifice-pitch bracket within which the experiment localises transition. Nothing is calibrated on this set.

Fig. 41. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the $\pm 0.025c$ orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred	err_c	err_pct	within_bracket	degenerate	burst	le_sep	x_sep_c	mechanism
1000 000.0	upper	1.35	6.483	1.350 3	0.175	0.203	0.0 28	16. 1	False	False	False	False		TS-natural
1000 000.0	upper	1.17 4	5.036	1.174 3	0.275	0.276	0.0 01	0.4	True	False	False	False		TS-natural
1000 000.0	upper	0.97 6	3.413	0.976 2	0.375	0.418	0.0 43	11. 5	False	False	False	False	0.33 5	separation
1000 000.0	upper	0.76 5	1.687	0.765	0.425	0.439	0.0 14	3.4	True	False	False	False	0.35 6	separation
1000 000.0	upper	0.41 9	- 1.132	0.418 7	0.475	0.461	- 0.0 14	-3.0	True	False	False	False	0.38 3	separation
1000 000.0	upper	- 0.03 9	- 4.846	- 0.038 7	0.575	0.504	- 0.0 71	- 12. 4	False	False	False	False		TS-natural
1000 000.0	lower	1.26 2	5.759	1.262 3	0.675	0.691	0.0 16	2.4	True	False	False	False	0.58 2	separation
1000 000.0	lower	1.07 5	4.224	1.075 3	0.675	0.704	0.0 29	4.4	False	False	False	False	0.58 2	separation
1000 000.0	lower	0.87 4	2.578	0.874 1	0.675	0.698	0.0 23	3.4	True	False	False	False	0.57 5	separation
1000 000.0	lower	0.65 1	0.757	0.650 8	0.675	0.698	0.0 23	3.4	True	False	False	False	0.56 8	separation
1000 000.0	lower	0.19 7	- 2.935	0.196 8	0.575	0.425	- 0.1 5	- 26. 0	False	False	False	False		TS-natural
2000 000.0	upper	1.46 6	7.438	1.466 1	0.075	0.124	0.0 49	66. 0	False	False	False	False		TS-natural
2000 000.0	upper	1.08 9	4.339	1.089 3	0.225	0.257	0.0 32	14. 3	False	False	False	False		TS-natural
2000 000.0	upper	0.88 3	2.652	0.883 1	0.325	0.335	0.0 1	3.2	True	False	False	False		TS-natural
2000 000.0	upper	0.55 1	- 0.058	0.550 7	0.425	0.425	0.0	0.1	True	False	False	False	0.36 9	separation
2000 000.0	upper	0.19 8	- 2.927	0.197 8	0.475	0.461	- 0.0 14	-3.0	True	False	False	False	0.40 4	separation
2000 000.0	upper	- 0.15 6	- 5.791	- 0.155 2	0.525	0.496	- 0.0 29	-5.4	False	False	False	False		TS-natural
2000 000.0	upper	- 0.49 1	- 8.514	- 0.491	0.625	0.589	- 0.0 36	-5.8	False	False	False	False		TS-natural
2000 000.0	upper	- 0.80 7	- 11.07 7	- 0.806 9	0.825	1.0	0.1 75	21. 2	False	True	True	False	0.94 7	none(laminar)
2000 000.0	lower	1.46 9	7.463	1.469 1	0.675	0.658	- 0.0 17	-2.5	True	False	False	False	0.58 9	separation
2000 000.0	lower	1.29 4	6.022	1.294 3	0.675	0.665	- 0.0 1	-1.5	True	False	False	False	0.58 9	separation
2000 000.0	lower	1.09 3	4.372	1.093 3	0.675	0.658	- 0.0 17	-2.5	True	False	False	False	0.58 2	separation
2000 000.0	lower	0.66 3	0.855	0.662 8	0.625	0.644	0.0 19	3.1	True	False	False	False	0.56 8	separation

2000 000.0	lower	0.43 4	-1.01	0.433 7	0.625	0.553	- 0.0 72	- 11. 4	False	False	Fal se	Fal se		TS- natural
3000 000.0	upper	1.09 4	4.38	1.094 3	0.175	0.227	0.0 52	29. 5	False	False	Fal se	Fal se		TS- natural
3000 000.0	upper	0.89	2.709	0.890 1	0.275	0.309	0.0 34	12. 2	False	False	Fal se	Fal se		TS- natural
3000 000.0	upper	0.66 2	0.847	0.661 8	0.375	0.404	0.0 29	7.8	False	False	Fal se	Fal se	0.36 3	separat ion
3000 000.0	upper	0.20 2	- 2.894	0.201 8	0.425	0.447	0.0 22	5.1	True	False	Fal se	Fal se	0.40 4	separat ion
3000 000.0	lower	1.62 3	8.733	1.622 7	0.675	0.651	- 0.0 24	-3.5	True	False	Fal se	Fal se	0.59 6	separat ion
3000 000.0	lower	1.47 2	7.487	1.472 1	0.675	0.644	- 0.0 31	-4.6	False	False	Fal se	Fal se	0.58 9	separat ion

Table 22. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled).

(table sampled to 30 rows; full data in 06_validation/aerofoil_nlf0416.csv)

11.3 Swept wings — the cross-flow branch

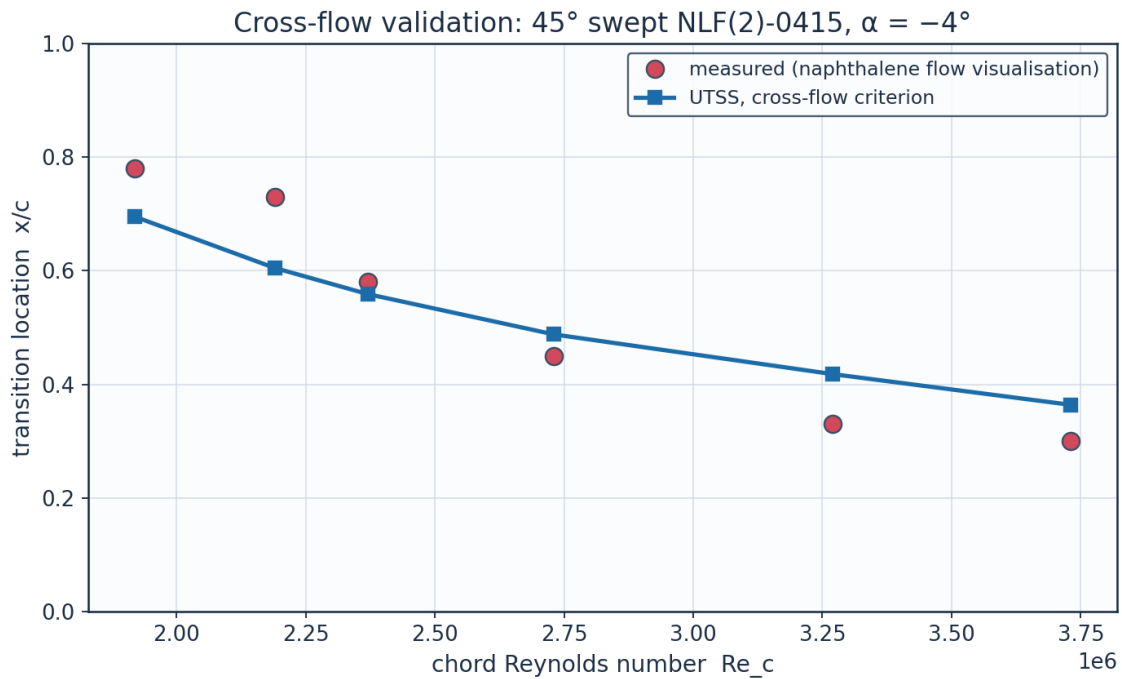
The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 14.7 % at the frozen constant $C1 = 150$. On the independent set that same constant gives 51.7 %, and $C1 = 200$ gives 18.4 % — so the criterion has the right functional form on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only $C1 = 150$ understates what the criterion does here, and reporting only $C1 = 200$ would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away. Nothing else in the model differs between the two columns.

What each experiment requires of the criterion is measured rather than asserted (Table 25). The march is run with every branch disabled so that it reaches the measured transition station, and the criterion is evaluated there. Dagenhart & Saric require a critical $Re_{\theta 2}$ of 169 with a 21.7 % coefficient of variation over six chord Reynolds numbers; Boltz et al. require 234 with 4.0 % over four sweep angles and a factor of three in chord Reynolds number. Each facility is therefore internally consistent — the second markedly so — and the two differ by 38 %. That is the shape of a receptivity difference rather than of a criterion with the wrong form: stationary cross-flow vortices are seeded by leading-edge roughness, and neither report documents the surface finish. Two attempts to close the gap fail and are recorded rather than dropped. Replacing the constant surrogate by the exact Falkner-Skan-Cooke factor $K(\lambda)$ makes matters worse, taking the pooled coefficient of variation from 22 to 86 % and inverting the ratio between the two sets. Giving the cross-flow branch its own amplification threshold, separate from the one Mack's relation supplies for Tollmien-Schlichting waves — which is defensible, since a stationary cross-flow vortex is not seeded by free-stream turbulence — moves the independent set only from 55 to 51 % as that threshold is taken from $N = 2$ to $N = 12$, with $C1$ refitted on the calibration set at each value. The reason is that once $Re_{\theta 2}$ exceeds $C1$ the amplification builds so quickly that

the threshold is nearly redundant with C1 itself, so the two constants cannot be separated by these data.

Re_c	x_tr_c_exp	x_tr_c_pred	err_pct	Re_theta_at_onset	mechanism
1920000.0	0.78	0.695	-10.9	667.0	crossflow
2190000.0	0.73	0.605	-17.1	662.5	crossflow
2370000.0	0.58	0.559	-3.7	662.7	crossflow
2730000.0	0.45	0.488	8.5	664.5	crossflow
3270000.0	0.33	0.418	26.6	670.7	crossflow
3730000.0	0.3	0.364	21.4	665.6	crossflow

Table 23. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).

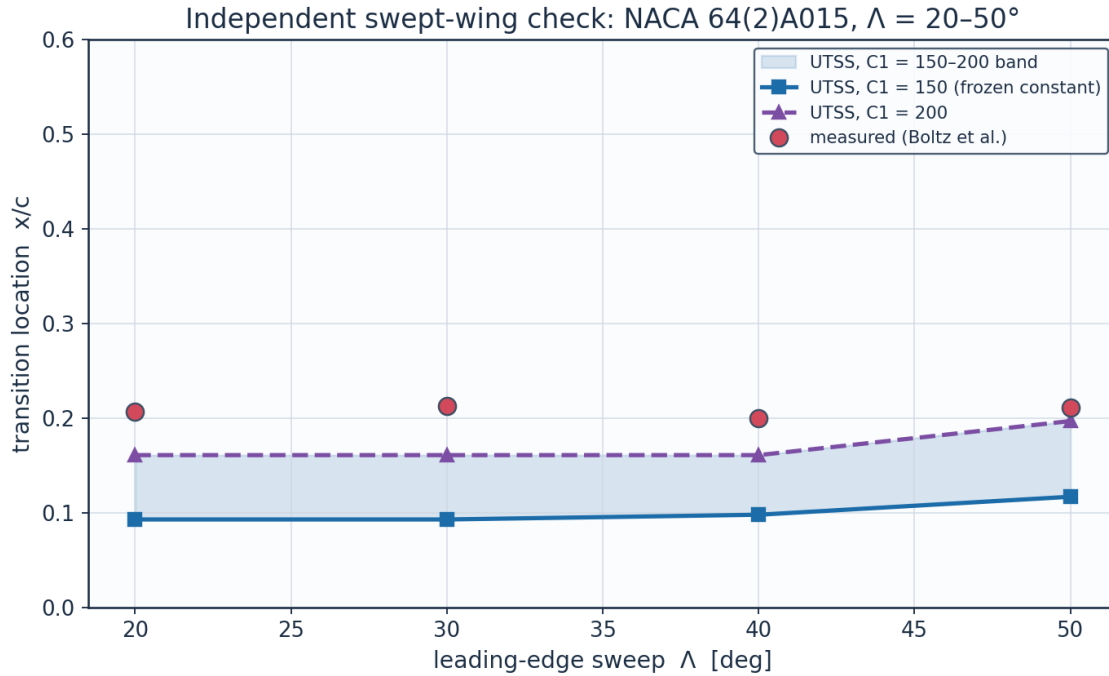


Source: Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wi...

Fig. 42. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	mechanism	x_tr_c_pred_C1_200	err_pct_C1_200
20.0	0.0	2700000.0	0.207	0.093	-55.0	crossflow	0.161	-22.4
30.0	0.0	1500000.0	0.213	0.093	-56.2	crossflow	0.161	-24.6
40.0	0.0	1200000.0	0.2	0.098	-51.1	crossflow	0.161	-19.7
50.0	0.0	9500000.0	0.211	0.117	-44.4	crossflow	0.197	-6.8

Table 24. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here.



Nothing is calibrated on this set. Source: Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boun...

Fig. 43. Transition location against sweep angle, NACA 64(2)A015, with the $C_1 = 150\text{--}200$ band shaded.

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

dataset	criterion	mean_critical_value	coeff_of_variation_pct	n_points
Dagenhart & Saric (calibration)	surrogate $Re_{\theta 2}$	168.9	21.7	6
Dagenhart & Saric (calibration)	exact Falkner-Skan-Cooke Re_{cf}	456.9	63.0	6
Boltz et al. (independent)	surrogate $Re_{\theta 2}$	233.8	4.0	4
Boltz et al. (independent)	exact Falkner-Skan-Cooke Re_{cf}	117.8	4.8	4
Both facilities pooled	surrogate $Re_{\theta 2}$	194.9	22.1	10
Both facilities pooled	exact Falkner-Skan-Cooke Re_{cf}	321.3	86.5	10

Table 25. What each swept-wing experiment requires of the cross-flow criterion, evaluated at the measured transition station (crossflow_criticals_summary.csv).

dataset	sweep_deg	Re_c	x_{tr_c} measured	$Re_{\theta 2}$	$Re_{\theta 2}$ surrogate	Re_{cf} exact_FSC
Dagenhart & Saric (calibration)	45.0	1920000.0	0.78	1041.7	244.8	971.7
Dagenhart & Saric (calibration)	45.0	2190000.0	0.73	769.7	180.9	718.0
Dagenhart & Saric (calibration)	45.0	2370000.0	0.58	675.0	158.6	351.8

Dagenhart & Saric (calibration)	45.0	2730000.0	0.45	637.1	149.7	275.8
Dagenhart & Saric (calibration)	45.0	3270000.0	0.33	590.7	138.8	215.0
Dagenhart & Saric (calibration)	45.0	3730000.0	0.3	598.6	140.7	209.1
Boltz et al. (independent)	20.0	27000000.0	0.207	1570.9	237.3	119.8
Boltz et al. (independent)	30.0	15000000.0	0.213	1189.0	242.0	119.6
Boltz et al. (independent)	40.0	12000000.0	0.2	1028.0	237.9	123.4
Boltz et al. (independent)	50.0	9500000.0	0.211	941.4	217.9	108.4

Table 26. Point-by-point cross-flow criterion values at the measured transition stations (crossflow_criticals.csv).

11.5 Ablations — what each element of the formulation is worth

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_pct
no bubble closure	5.6	0.0607	19	86	22.1
one-equation laminar march	-4.7	0.0388	44	86	51.2
Drela-Giles envelope	-6.6	0.0451	39	86	45.3
full model	5.6	0.0373	52	86	60.5

Table 27. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted, and so returns 8.18 for any stream quieter than 0.08 %. The SAME constants reproduce every validation dataset — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

11.6 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA

		J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCOFTAC T3A- flat plate	Roach & Brierley (1990), ERCOFTAC T3A- (Tu=0.87%); ERCOFTAC Classic Collection Case 020.
Validation	ERCOFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification database	Orr (1907); Sommerfeld (1908); Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10), 1347.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 28. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 29. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as $x/c = R_{trans}/R$.

Table 30. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
McGhee et al. NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-2328, Fig. 25.	86	0.03	0.05

Table 31. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at $Re_{\theta} = 200$ against the accepted 200.5.
- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_{crit} used elsewhere, so the length scales with the disturbance environment.

- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 44 to 52, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset Re_{θ_t} predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -28.8 % on T3C4, +5.6 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.037 chord, and 52 of the 86 predictions fall inside the $\pm 0.025c$ bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry $\rightarrow$ mesh $\rightarrow$ setup $\rightarrow$ solution $\rightarrow$ post-processing $\rightarrow$ validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: 58 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 58 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 50 % larger than the first. And the method is an attached-flow formulation with an incidence envelope of about ± 8.5 deg on a real section; beyond that the laminar march separates near the leading edge and the result should not be relied on.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 107 (CSV: 46, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG

05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/crossflow_criticals.csv	CSV
06_validation/crossflow_criticals_summary.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG

06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall normal reconstruction stack (no volume mesh is generated)	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H^* , I and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N , one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	$C1$ on $Re_{\theta 2}$, closed by an amplification integral
Intermittency closure	Narasimha universal (γ)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann C_f
Laminar separation criterion	Thwaites $\lambda \leq -0.09$
Turbulent separation flag	$H > 2.6$
Drag integration	Squire-Young far-wake, evaluated at $0.98c$
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	$1e-10$ (direct solve)
Wall y^+ target	~ 1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00438	-
Section Cd (counts)	43.8	counts
Section L/D	118.0	-
Wing lift (3-D estimate, $C_L = 0.90 C_{L_i}$)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.